

Research Programme Overview

Non-Bank Financial Intermediation and Systemic Risk

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Abstract

This document provides a detailed research cookbook for a four-project programme studying how non-bank financial intermediation (NBFI) creates, amplifies, and transmits systemic risk. For each project, every analytical module specifies: (1) variables—exact names, definitions, sources, frequency, and level of aggregation; (2) model specification—full equation with every term defined; (3) estimation—method, software, key parameters; (4) output—what you get and how to interpret it; and (5) module linkages—how each module’s output feeds into the next. The unifying theme is that NBFI risks are fundamentally nonlinear: largely invisible in normal times but activating powerfully during stress.

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1 Research Programme Overview: Non-Bank Financial Intermediation and Systemic Risk

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This document is a **detailed research cookbook** for the four-project programme. For each project, every analytical module specifies:

1. **Variables** — exact names, definitions, sources, frequency, and level of aggregation
2. **Model specification** — full equation with every term defined
3. **Estimation** — method, software, key parameters (lags, quantiles, etc.)
4. **Output** — what you get and how to interpret it
5. **Module linkages** — how each module's output feeds into the next

Unifying theme: NBFI risks are fundamentally nonlinear — largely invisible in normal times but activating powerfully during stress. Standard mean-based methods miss these dynamics. All four projects employ quantile-specific methods to capture tail amplification and asymmetric contagion.

Project	Title	Focus
P1	The Shadow Leverage Map	Hidden NBFI leverage from repo & derivatives
P2	Mapping the Channels	NBFI amplification of monetary policy transmission
P3	Contagion Across Borders	Cross-border NBFI spillovers via the GFC
P4	FX Hedging as a Contagion Channel	FX hedging costs as a transmission mechanism

1.1 How to Read This Document

Each project is organised into **numbered modules**. Every module follows this template:

Purpose — *what economic question this module answers*

Unit of observation — *e.g., "sub-sector \$\$\$ × quarter \$t\$"*

Input variables — table listing every variable entering the model, its definition, source, frequency, and level of aggregation

Model specification — the full equation with all terms defined

Estimation details — how the model is estimated (OLS, quantile regression, MLE, simulation, etc.), plus key parameters

Output — what the function returns and how to interpret the results

→ **Link to next module** — what this module's output feeds into

At the end of each project there is a **Module Linkage Map** showing the pipeline flow.

2 Project 1: The Shadow Leverage Map

Subtitle: Tracing Hidden Non-Bank Exposures Through Repo and Derivatives Data

2.1 Motivation

NBFIs now account for nearly half of global financial assets, yet their leverage remains largely invisible to regulators. Unlike banks, NBFIs are not subject to standardised capital or leverage ratio disclosure, creating a systemic blind spot. This project develops a *Shadow Leverage Map* — a prototype analytical framework that reverse-engineers the leverage embedded in NBFI balance sheets using publicly available market data from repo markets (OFR, ECB MMSR), derivatives reporting (DTCC), and FSB monitoring data.

2.2 Research Questions

RQ1. Can NBFI leverage be reverse-engineered from observable market data (repo volumes, derivatives notional, equity proxies)?

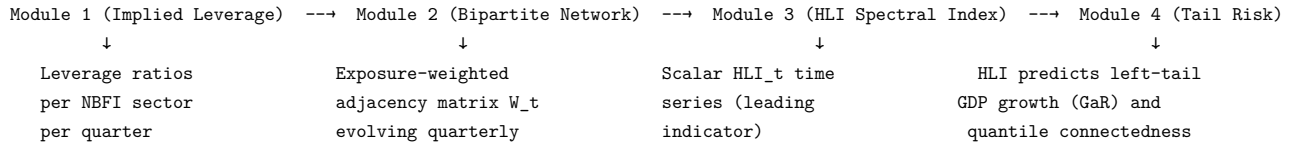
RQ2. How are banks and NBFIs interconnected through repo and derivatives exposures, and how do these interconnections evolve over time and during stress?

RQ3. Can a spectral measure of network structure — the Hidden Leverage Index (HLI), based on the largest eigenvalue of the exposure-weighted adjacency matrix — serve as a predictor of systemic events?

RQ4. Does hidden leverage amplify tail risk during stress? Is the bank-NBFI system more interconnected in the tails (quantile connectedness), and does high hidden leverage predict worse left-tail growth (Growth-at-Risk)?

2.3 Empirical Strategy

Project 1 has **four modules** that build on each other in sequence:



2.3.1 Module 1: Implied Leverage Estimation

Purpose: Reverse-engineer the economic leverage embedded in NBFI balance sheets — leverage that regulatory ratios do not capture — using observable market data.

Unit of observation: NBFI sub-sector $\$s\$ \times$ quarter $\$t\$$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Equity proxy	$E_{s,t}$	Total net asset value (equity) of NBFI sub-sector $\$s\$$	FSB Global Monitoring Report; Fed Z.1 Flow of Funds	Annual (FSB); Quarterly (Z.1)	Sub-sector aggregate
Repo borrowing	$R_{s,t}$	Total repo borrowing by sub-sector $\$s\$$	OFR US Repo (SOFR data); ECB MMSR	Daily (aggregated to quarterly)	Sub-sector aggregate
Derivatives notional	$D_{s,t}$	Gross derivatives notional outstanding for sub-sector $\$s\$$	DTCC Swap Data Repository	Weekly (aggregated to quarterly)	Sub-sector aggregate
Delta-equivalence factor	δ	Scaling factor converting gross notional to economic exposure	Calibrated from BIS Triennial Survey	Fixed parameter	Global

Sub-sectors $s \in \{\text{hedge funds, MMFs, bond mutual funds, equity mutual funds, ETFs, broker-dealers, insurance companies, pension funds, finance companies}\}$.

Model specification:

$$\text{Implied Leverage}_{s,t} = \frac{E_{s,t} + R_{s,t} + \delta \cdot D_{s,t}}{E_{s,t}}$$

Worked example: A hedge fund sector with $E = \$10\text{B}$ equity, $R = \$40\text{B}$ repo borrowing, and $D = \$500\text{B}$ IRS notional (with $\delta = 0.05$) has implied leverage of $(10 + 40 + 0.05 \times 500) / 10 = 7.5\times$.

Estimation details:

- This is a *constructed measure*, not an econometric model — no estimation required.
- The key calibration choice is δ . Baseline: $\delta = 0.05$ (from BIS survey data on delta-equivalence of IRS portfolios). Robustness: $\delta \in [0.02, 0.10]$.
- For sub-sectors where direct repo data is unavailable, repo borrowing is proxied by total short-term funding minus equity, using Flow of Funds (Z.1) liability breakdowns.

Output: A panel of implied leverage ratios: $\text{Lev}_{s,t}$ for each sub-sector s and each quarter t (2013Q1–2024Q4). This is a single number per sub-sector per quarter — the “true” economic leverage.

→ **Link to Module 2:** The implied leverage ratios serve as *edge weights* in the bipartite network. Higher leverage for sub-sector s means the edges from banks to s carry more systemic weight (because a given $\$$ exposure to a highly-leveraged NBFI poses more fire-sale risk).

2.3.2 Module 2: Bipartite Bank-NBFI Network

Purpose: Map the bilateral exposure network between G-SIB banks and NBFI sub-sectors, weighted by both exposure amounts and the implied leverage from Module 1.

Geographic Dimension of Exposures The economically correct unit of observation is:

$\text{Exp}_{n,i,j,m,t}$ = exposure of bank i in country n to NBFI sub-sector j in country m at time t

However, bilateral bank-by-NBFI-by-country data is generally unavailable. The realistic data structure is **semi-bilateral**: for each NBFI sub-sector j in country n , we observe total cross-border exposure to the rest of the world, $\text{Exp}_{j,n,t}^{\text{RoW}}$, but not the specific (n,m) counterparty breakdown at the sub-sector level. This means:

- **Domestic exposures:** Where bilateral data exists (e.g., US via OFR, EU via ECB MMSR), we can construct full bank-by-NBFI matrices within a jurisdiction.

- **Cross-border exposures:** Available only as (sub-sector j , country n) $\rightarrow$ rest-of-world aggregates (from BIS Locational Banking Statistics, which report bank claims on/liabilities to non-bank financial institutions by country, but not by NBFI sub-type across borders).

In practice, W_t is constructed as a **two-tier structure**:

1. **Within-jurisdiction:** Full (i,j) bilateral matrix where OFR/ECB data permits.
2. **Cross-border:** $(j,n) \rightarrow$ RoW aggregate exposures, allocated proportionally using BIS country-level banking statistics as weights.

This is an honest limitation. We can still compute system-wide amplification (λ_1), but the cross-border contagion paths are identified at the country level rather than the bank-NBFI pair level. The geographic decomposition connects directly to Project 3 (cross-border contagion).

Input variables

Variable	Symbol	Definition	Source	Frequency	Level
Bilateral exposure (domestic)	$\text{Exp}_{ij,t}$	Bank i 's total exposure to NBFI sub-sector j (within jurisdiction)	OFR bilateral repo data (US); ECB MMSR (EU)	Quarterly	Bank $\times$ NBFI sub-sector
Cross-border exposure	$\text{Exp}_{j,n,t}^{\text{RoW}}$	Total exposure of NBFI sub-sector j in country n to rest-of-world	BIS Locational Banking Stats	Quarterly	Sub-sector $\times$ country
Implied leverage	$\text{Lev}_{j,t}$	From Module 1	Module 1 output	Quarterly	NBFI sub-sector
Institution metadata	—	Bank names, NBFI sub-sector classification, country	FSB G-SIB list; own classification	Static	Institution

Network construction The network is a **directed weighted bipartite graph** $G_t = (V_B \cup V_N, E_t)$ where V_B = G-SIB banks, V_N = NBFI sub-sectors, and E_t = directed edges representing credit and funding relationships.

Construction of the Adjacency Matrix W_t The **exposure-weighted adjacency matrix** is:

$$W_{ij,t}^{\text{credit}} = \frac{\text{Exp}_{ij,t} \times \text{Lev}_{j,t}}{\sum_k \text{Exp}_{ik,t} \times \text{Lev}_{k,t}}$$

Why multiply by leverage? The logic is a **fire-sale argument**. Consider bank i that has

lent \$100M to NBFI sub-sector j via repo. If j is leveraged $2\times$, a 10% asset price decline wipes out 20% of its equity — painful but survivable. If j is leveraged $10\times$, the same 10% decline wipes out 100% of equity — triggering forced liquidation, fire sales, and potential default. The loss that cascades back to bank i is therefore proportional not just to the raw exposure $\text{Exp}_{ij,t}$ but to the *leverage-adjusted exposure* $\text{Exp}_{ij,t} \times \text{Lev}_{j,t}$. Leverage acts as an amplifier: the same dollar of exposure to a highly-leveraged counterparty poses far more systemic risk than to a low-leverage one.

The row-normalisation (dividing by $\sum_k$) ensures each row sums to 1, giving the *share* of bank i 's leverage-weighted risk that comes from each NBFI sub-sector.

Two Directions of Risk: Credit vs. Funding The matrix W_t^{credit} (dimension $n_B \times n_N$) captures only one direction — **credit/fire-sale risk** (bank's loss from NBFI distress). There is a second, equally important direction — **funding risk** (NBFI's loss from bank funding withdrawal). The funding matrix W_t^{funding} has dimension $n_N \times n_B$, with (j,i) -th entry:

$$W_{ji,t}^{\text{funding}} = \frac{\text{Exp}_{ji,t} \times \text{Lev}_{i,t}}{\sum_k \text{Exp}_{jk,t} \times \text{Lev}_{k,t}}$$

This captures: NBFI sub-sector j depends on bank i for repo funding or credit lines. If bank i is highly leveraged and its own VaR constraint binds, it pulls back funding (raises haircuts, calls margin, refuses to roll repo). The severity depends on how leveraged bank i is — more leveraged banks are more likely to pull funding when stressed.

The Combined System Matrix For the spectral analysis (Module 3), we need a **single** matrix that captures the feedback loop between credit and funding risk: bank pulls funding $\rightarrow$ NBFI fire-sells $\rightarrow$ bank takes credit loss $\rightarrow$ bank pulls more funding. Two separate matrices would miss this circularity.

We combine both into a single $(n_B + n_N) \times (n_B + n_N)$ block matrix:

$$\mathcal{W}_t = \begin{pmatrix} 0 & W_t^{\text{credit}} \\ W_t^{\text{funding}} & 0 \end{pmatrix}$$

where:

- The upper-right block W_t^{credit} ($n_B \times n_N$): bank i 's credit risk exposure to NBFI j
- The lower-left block W_t^{funding} ($n_N \times n_B$): NBFI j 's funding risk from bank i
- The diagonal blocks are zero (within-bank and within-NBFI linkages are not modelled here)

The spectral radius $\lambda_1(\mathcal{W}_t)$ of this combined matrix captures the **total amplification potential** from both channels feeding into each other. A high λ_1 means the credit-funding feedback loop is tightly wound — NBFI fire sales hurt banks, which pull funding, which triggers more fire sales.

NBFI Counterparty Decomposition A critical refinement: the systemic risk implications of NBFI activity depend fundamentally on **who the NBFI transacts with**. An NBFI sub-sector’s balance sheet should be decomposed by counterparty:

Domestic banks	Foreign counterparties	Domestic private sector
NBFI assets (lends to)	Repo lending, deposits, bank bonds → NBFI is <i>funding source</i> for banks	Cross-border portfolio holdings, FX swaps → NBFI transmits shocks <i>outward</i>
NBFI liabilities (borrows from)	Repo borrowing, credit lines, prime brokerage → Bank exposed to NBFI <i>credit risk</i>	Cross-border funding, FX swap borrowing → NBFI <i>imports</i> foreign funding shocks

Each cell creates a different contagion channel with different speed, amplification, and policy implications. A country where NBFIs primarily borrow from domestic banks and lend to the private sector (credit intermediation model) has a completely different risk profile from one where NBFIs borrow from foreigners and trade derivatives with domestic banks (market intermediation model).

Data source: “From-whom-to-whom” financial accounts — Fed Z.1 (US), ECB Financial Accounts (euro area), OECD institutional sector accounts (broader coverage).

Implication for the adjacency matrix: Where from-whom-to-whom data is available, W_t can be disaggregated by counterparty type, enabling separate spectral analyses for credit-intermediation vs. market-intermediation risk channels.

Estimation details

- No econometric estimation — this is a data construction step.
- Network is built using `networkx` (Python). The function `build_exposure_network()` creates a `DiGraph` from the exposure panel.
- Centrality measures computed: PageRank, eigenvector centrality, betweenness centrality, in/out-strength (weighted degree).
- The network evolves quarterly from 2015Q1 to 2024Q4 (limited by BIS sectoral breakdown availability; extended to 2013 where OFR bilateral data permits).

Output

- A time series of adjacency matrices $W_t^{\text{credit}} (n_B \times n_N)$ and $W_t^{\text{funding}} (n_N \times n_B)$, one pair per quarter.
- Combined system matrix $\mathcal{W}_t ((n_B + n_N) \times (n_B + n_N))$ block matrix) for spectral analysis.
- Centrality rankings for each node at each date.
- Network summary statistics over time: density, total exposure, HHI (concentration), reciprocity.

- Decomposition of cross-border vs. domestic exposure shares over time.

→ **Link to Module 3:** The combined system matrix $\mathcal{W}_t$ is the direct input to the spectral analysis. The HLI is the largest eigenvalue of $\mathcal{W}_t$.

2.3.3 Module 3: Hidden Leverage Index (Spectral)

Purpose: Collapse the bipartite network into a single scalar measure of systemic amplification potential — the Hidden Leverage Index (HLI) — using the spectral radius of the exposure matrix.

Unit of observation: Aggregate system-level measure $\times$ quarter \$t\$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Combined adjacency matrix	$\mathcal{W}_t$	Combined credit + funding block matrix from Module 2	Module 2 output	Quarterly	System-level

Model specification:

$$\text{HLI}_t = \lambda_1(\mathcal{W}_t)$$

where $\lambda_1(\mathcal{W}_t)$ is the **largest eigenvalue** (spectral radius) of the combined system matrix $\mathcal{W}_t$.

Economic intuition — network amplification, not PCA (Acemoglu, Ozdaglar & Tahbaz-Salehi 2015): The spectral radius is *not* a statistical summary like PCA's first component — it is a **structural measure of the network's amplification capacity**. The logic is a cascade argument:

Consider a shock ε hitting one node. Through the network, losses propagate: node j 's distress hits bank i with weight $\mathcal{W}_{ij}$, then i 's distress hits others with weight $\mathcal{W}_{ki}$, and so on. After n rounds, total losses are:

$$\text{Total loss} \propto \varepsilon + \mathcal{W}\varepsilon + \mathcal{W}^2\varepsilon + \mathcal{W}^3\varepsilon + \dots = (I - \mathcal{W})^{-1}\varepsilon$$

This geometric series converges if and only if $\lambda_1(\mathcal{W}) < 1$. The **rate of amplification** is governed by λ_1 :

- λ_1 close to 0: shocks dissipate quickly. The network is "loose."

- λ_1 close to 1: shocks amplify through many rounds before dying out. The network is "tightly wound" — approaching the instability threshold.

The **eigenvector** associated with λ_1 also matters — it identifies **which nodes are most central to the amplification**. If the leading eigenvector loads heavily on hedge funds, that is where systemic risk concentrates.

Estimation details:

- Computed via `numpy.linalg.eigvals` on $\mathcal{W}_t$ at each quarter.
- No estimation — this is a deterministic transformation of $\mathcal{W}_t$.
- Robustness: also compute the second eigenvalue λ_2 (captures secondary amplification channels) and the spectral gap $\lambda_1 - \lambda_2$ (captures how "dominant" the main amplification channel is).
- Separate spectral analysis on W^{credit} vs. W^{funding} to identify which direction of risk dominates amplification.

Output: A quarterly time series $\{\text{HLI}_t\}_{t=2013Q1}^{2024Q4}$ — a single number per quarter measuring systemic amplification potential.

Validation: The HLI should spike *before* known stress episodes:

- March 2020 (COVID dash-for-cash)
- September 2022 (UK gilt/LDI crisis)
- March 2023 (SVB failure)

If it does, this confirms its value as a leading indicator.

→ **Link to Module 4:** The HLI time series enters Module 4 as a predictor in tail risk regressions (quantile connectedness and Growth-at-Risk).

2.3.4 Module 4: Tail Risk Econometrics

Module 4 has **three sub-analyses**, all using the HLI from Module 3 as a key variable.

Module 4a: Quantile Connectedness (Bank-NBFI System) **Purpose:** Test whether the bank-NBFI system is more interconnected in the tails (stress) than at the median (normal times).

Unit of observation: System of k return series $\times$ week t .

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Bank sector re- turns	$r_{i,t}^B$	Weekly log- returns for G-SIB banks (or bank sector index)	Bloomberg / Refinitiv	Weekly	Institution or sector
NBFI sector re- turns	$r_{j,t}^N$	Weekly log- returns for NBFI sub- sector indices (HF index, MMF index, insurance index, etc.)	Bloomberg (HFRX, etc.)	Weekly	Sub-sector index

System: A vector $Y_t = (r_1^B, \dots, r_m^B, r_1^N, \dots, r_n^N)'$ of $k = m + n$ return series (e.g., 5 bank sector returns + 5 NBFI sub-sector returns = 10-variable system).

Model specification — Quantile VAR at quantile τ :

$$Q_\tau(y_{i,t} \mid Y_{t-1}, \dots, Y_{t-p}) = c_i(\tau) + \sum_{j=1}^k \sum_{l=1}^p \beta_{ij}^{(l)}(\tau) y_{j,t-l}$$

This is a standard VAR except that *every coefficient depends on the quantile τ* . At $\tau = 0.05$, you are modelling the left tail; at $\tau = 0.50$, the median; at $\tau = 0.95$, the right tail.

Estimation details:

- Estimated **equation by equation** using `statsmodels.QuantReg` (linear quantile regression, Koenker & Bassett 1978).
- For each equation i : the dependent variable is $y_{i,t}$; the regressors are p lags of all k variables plus an intercept.
- Lags: $p = 4$ (quarterly data) or $p = 2$ (weekly data). Selected via BIC at each quantile.
- Quantiles estimated: $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$.
- From the estimated QVAR, compute the **Generalised Forecast Error Variance Decomposition (GFEVD)** at horizon $h = 10$:
- Uses the companion matrix to compute MA representation.
- GFEVD entry $\theta_{ij}(\tau)$ = share of variable i 's h -step forecast error variance attributable to shocks in variable j , at quantile τ .

- Normalised so rows sum to 100%.

Connectedness measures (Ando, Greenwood-Nimmo & Shin 2022):

- **Total connectedness:** $C(\tau) = \frac{1}{k} \sum_{i \neq j} \theta_{ij}(\tau) \times 100\%$
- **To-others:** $TO_i(\tau) = \sum_{j \neq i} \theta_{ji}(\tau)$ — how much i transmits to the system
- **From-others:** $FROM_i(\tau) = \sum_{j \neq i} \theta_{ij}(\tau)$ — how much i receives from the system
- **Net:** $NET_i(\tau) = TO_i - FROM_i$ — net transmitter (>0) or net receiver (<0)

Output:

1. A $k \times k$ GFEVD matrix $\Theta(\tau)$ at each quantile — shows pairwise spillover direction and magnitude.
2. Total connectedness at each quantile: $C(0.05)$, $C(0.50)$, $C(0.95)$.
3. Directional spillovers: which sectors are net transmitters vs. receivers of risk, and how this changes between median and tails.

Key test: Is $C(0.05) \gg C(0.50)$? If total connectedness at the 5th percentile is much higher than at the median, the system is asymmetrically interconnected — more contagious during stress.

Rolling version: Estimated on a rolling window (60 weeks), stepping 1 week at a time. Produces a time series of $C_t(0.05)$ and $C_t(0.50)$ that tracks how tail connectedness evolves through crisis episodes.

Module 4b: Tail-Risk Amplification (Quantile Regression with HLI Interaction)

Purpose: Test whether high hidden leverage amplifies left-tail outcomes — i.e., does the HLI predict *worse* downside risk during stress?

Unit of observation: Quarter t (aggregate time series).

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Outcome variable	y_t	System-wide equity return or financial conditions index	Bloomberg / FRED	Quarterly	Aggregate

Variable	Symbol	Definition	Source	Frequency	Level
Stress indicator	Stress_t	VIX level, or dummy for VIX $\$>\$$ 75th percentile	CBOE (via FRED)	Quarterly	Aggregate
Hidden Leverage Index	HLI_t	From Module 3	Module 3 output	Quarterly	Aggregate
Controls	X_t	Term spread (10Y-2Y), credit spread (BAA-AAA), GDP growth	FRED	Quarterly	Aggregate

Model specification — Quantile regression at $\tau = 0.05$:

$$Q_\tau(y_t) = \alpha(\tau) + \beta_1(\tau) \cdot \text{Stress}_t + \beta_2(\tau) \cdot \text{HLI}_{t-1} + \beta_3(\tau) \cdot (\text{Stress}_t \times \text{HLI}_{t-1}) + \gamma(\tau)' X_{t-1} + \varepsilon_t$$

Note: HLI is **lagged** (predictive, not contemporaneous) to address endogeneity.

Key parameter: $\beta_3(\tau = 0.05)$. If negative and significant: high hidden leverage makes left-tail outcomes *worse* during stress. If $\beta_3(\tau = 0.50) \approx 0$: no effect at the median — confirming this is a *tail-specific* amplification mechanism.

Estimation details:

- `statsmodels.QuantReg` at $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$.
- Standard errors: Kernel (Hall-Sheather bandwidth) or bootstrap (1000 replications).
- Formal interquantile test: Wald test of $H_0 : \beta_3(0.05) = \beta_3(0.50)$.

Output: Table of coefficients across quantiles, showing that the HLI $\times$ Stress interaction is large and negative at $\tau = 0.05$ but near zero at $\tau = 0.50$.

Module 4c: Growth-at-Risk (Adrian et al. 2019) **Purpose:** Test whether the HLI predicts worse left-tail GDP growth — extending the GaR framework with an NBFi-specific predictor.

Unit of observation: Quarter $\$t\$$ (aggregate time series). Potentially also country $\$c\$ \times$ quarter $\$t\$$ (panel).

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Future GDP growth	y_{t+h}	Real GDP growth at horizon h (1, 4, 8, 12 quarters ahead)	FRED (GDPC1)	Quarterly	Aggregate
Financial conditions	FCI_t	National Financial Conditions Index or composite (VIX + credit spread + term spread)	Chicago Fed NFCI; or constructed	Quarterly	Aggregate
Hidden Leverage Index	HLI_t	From Module 3	Module 3 output	Quarterly	Aggregate

Model specification — Quantile regression:

$$Q_\tau(y_{t+h}) = \alpha(\tau) + \beta_1(\tau) \cdot \text{FCI}_t + \beta_2(\tau) \cdot \text{HLI}_t + \varepsilon_t$$

Estimated at multiple horizons $h \in \{1, 2, 4, 8, 12\}$ quarters and multiple quantiles $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$

Key test: $\beta_2(\tau = 0.05) < 0$ and significant, but $\beta_2(\tau = 0.50) \approx 0$. This means high hidden leverage predicts worse left-tail growth outcomes but has no effect on median growth — confirming that shadow leverage is a *tail risk* phenomenon.

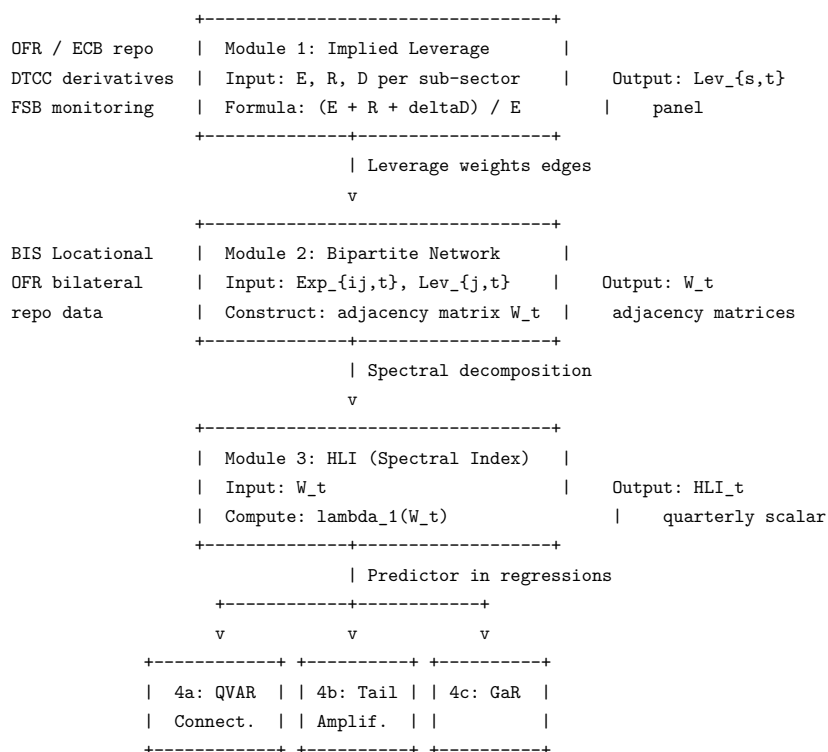
Estimation details:

- `statsmodels.QuantReg` with `growth_at_risk()` function from `src/econometrics/quantile_var.py`.
- Also estimate the "term structure of GaR": how the FCI and HLI effects change across horizons h .
- Extended sample: 2000Q1–2024Q4 using annual FSB data interpolated to quarterly frequency for the HLI.

Output:

1. **GaR term structure plot:** for each horizon h , the 5th percentile of predicted GDP growth, split by HLI high vs. HLI low.
2. **Coefficient table:** $\beta_2(\tau)$ across quantiles — showing the HLI matters only in the left tail.
3. **Fan chart:** predicted distribution of future GDP growth conditional on current FCI and HLI values.

2.3.5 Module Linkage Map (Project 1)



2.4 Expected Contributions

1. First implied leverage measure for NBFIs from publicly observable market data.
2. Hidden Leverage Index as a leading indicator — spikes before March 2020 and September 2022.
3. Shadow leverage amplifies left-tail risk (GaR) but has no effect at the median.
4. Quantile connectedness confirms asymmetric contagion in bank-NBFI networks.

2.5 Data Sources & Sample

Data Source	Variables	Frequency	Coverage
OFR US Repo (SOFR)	Repo volumes, rates, counterparty type	Daily	2014–present
ECB MMSR	Euro repo volumes by counterparty	Daily	2016–present
DTCC Swap Data Repository	IRS/CDS notional, counterparty type	Weekly	2013–present
FSB Global Monitoring Report	NBFI sector AUM, leverage proxies	Annual	2002–present

Data Source	Variables	Frequency	Coverage
BIS Locational Banking Stats	Cross-border bank-NBFI exposures	Quarterly	2000–present
FRED / ECB SDW	GDP, financial conditions, VIX, spreads	Mixed	2000–present

Sample period: 2013Q1–2024Q4 (post-DTCC reporting). Extended to 2000 for GaR regressions using FSB annual data.

Key stress episodes for validation: Taper Tantrum (May 2013), China devaluation (Aug 2015), Gilt crisis (Sep 2022), COVID-19 (Mar 2020), SVB failure (Mar 2023).

2.6 Literature Positioning

This project contributes to three strands:

1. **NBFI leverage measurement:** Extends Jiang, Matvos, Piskorski & Seru (2024) on hidden bank losses to the NBFI sector. Complements FSB (2023) Global Monitoring Report with a market-data-based approach.
2. **Spectral network measures:** Builds on Acemoglu, Ozdaglar & Tahbaz-Salehi (2015) and Greenwood, Landier & Thesmar (2015) on network-based contagion measures. The HLI adapts the spectral radius concept to a bipartite bank-NBFI setting.
3. **Growth-at-Risk:** Extends Adrian, Boyarchenko & Giannone (2019) by adding NBFI-specific predictors (hidden leverage) to the financial conditions $\rightarrow$ GDP growth-at-risk framework.

2.7 Identification & Robustness

- **Endogeneity of leverage:** Leverage is both a response to and a predictor of stress. We address this via (a) lagged HLI values, (b) Granger causality tests, and (c) instrumenting leverage with regulatory threshold dummies (e.g., margin call triggers).
- **Measurement error in implied leverage:** Delta-equivalence factor δ is calibrated from BIS survey data; robustness across $\delta \in [0.02, 0.10]$.
- **Network construction sensitivity:** Results tested across alternative edge-weighting schemes (gross vs net, bilateral vs multilateral netting).

3 Project 2: Mapping the Channels

Subtitle: Network Analysis of Non-Bank Amplification in Monetary Policy Transmission

3.1 Motivation

Building on Banerjee, Hofmann, Ng & Pinter (2025, BIS Bulletin 116), who document that other financial intermediaries (OFIs) amplify monetary policy transmission to long-term yields and credit spreads while insurance companies and pension funds (ICPFs) dampen it, we decompose this aggregate effect using granular network and connectedness methods. Their panel local-projection analysis reveals significant amplification but with wide confidence bands, leaving open: *which* OFI sub-sectors drive it, *through which* balance-sheet channels, and *under what* conditions.

3.2 Hypotheses

H1 (Heterogeneous amplification). Procyclicality beta is highest for hedge funds and leveraged investment funds, moderate for bond mutual funds and MMFs, and lowest for pension funds and insurance.

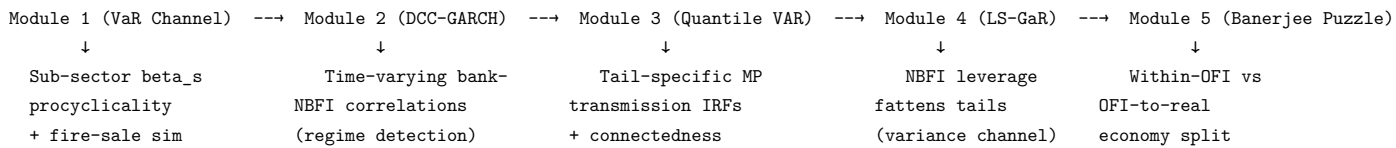
H2 (Nonlinear tail transmission). Quantile IRFs at $\tau = 0.05$ show $2\text{--}3\times$ stronger responses than at $\tau = 0.50$, especially for the policy rate $\rightarrow$ hedge fund returns link.

H3 (Variance amplification). NBFI leverage increases the conditional variance (not just the conditional mean) of future growth outcomes, generating fatter left tails.

H4 (Network containment). During the 2022–23 hiking cycle, within-OFI connectedness was high but OFI-to-real-economy connectedness remained low, explaining the resilience puzzle.

3.3 Empirical Strategy

Project 2 has **five modules**. Module 1 establishes the micro-foundation (VaR-constraint channel). Modules 2–3 provide the econometric core. Modules 4–5 address variance effects and the 2022–23 puzzle.



3.3.1 Module 1: VaR-Constraint Channel (Adrian & Shin 2010, 2014)

Economic logic Financial intermediaries subject to Value-at-Risk (VaR) constraints manage their balance sheets by targeting a maximum loss at a given confidence level. When asset prices rise, measured portfolio risk (VaR) falls mechanically, creating unused risk capacity. Intermediaries respond by expanding their balance sheets — buying more assets funded by short-term debt — so that leverage *rises* with total assets. This is the **procyclicality of leverage**: $\text{Corr}(\Delta \log \text{Leverage}, \Delta \log \text{Assets}) > 0$.

The reverse is more dangerous. When asset prices fall, VaR rises, breaching the constraint. Intermediaries must sell assets to restore compliance, which depresses prices further, triggering

more VaR breaches across interconnected institutions — the **fire-sale spiral**. Adrian & Shin (2010, 2014) document this mechanism for US broker-dealers. We extend it to the full NBFI universe and test whether the mechanism is heterogeneous across sub-sectors.

What we test

1. **Procyclicality by sub-sector (H1):** Is leverage procyclical for each NBFI type, and is the procyclicality coefficient β_s larger for leveraged sub-sectors (hedge funds) than for liability-driven ones (pension funds, insurance)?
2. **Fire-sale amplification:** Does adding NBFI sub-sectors to a bank-only fire-sale simulation produce larger aggregate losses, and which sub-sectors contribute most to the amplification?

Specification 1: Procyclicality of leverage (sub-sector panel) Unit of observation: NBFI sub-sector \$\$ in quarter \$t\$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Leverage	$Lev_{s,t}$	Total assets / equity (net asset value) of sub-sector \$\$	Fed Z.1 Flow of Funds; ECB Investment Fund Statistics	Quarterly	Sub-sector aggregate
Total assets	$A_{s,t}$	Total financial assets of sub-sector \$\$	Fed Z.1 (tables L.122–L.130)	Quarterly	Sub-sector aggregate
Equity	$E_{s,t}$	Net asset value (= assets - liabilities) of sub-sector \$\$	Fed Z.1	Quarterly	Sub-sector aggregate
VIX	VIX_t	CBOE Volatility Index (quarterly average of daily values)	FRED (VIX-CLS)	Daily → Quarterly avg	Aggregate
Term spread	—	10Y Treasury - 2Y Treasury yield	FRED (T10Y2Y)	Daily → Quarterly avg	Aggregate
Credit spread	—	BAA - AAA corporate bond spread	FRED (BAA - AAA)	Daily → Quarterly avg	Aggregate

Variable	Symbol	Definition	Source	Frequency	Level
GDP growth	—	Real GDP quarter-on-quarter growth (annualised)	FRED (GDPC1)	Quarterly	Aggregate

Sub-sectors $s \in \{\text{hedge funds, MMFs, bond mutual funds, equity mutual funds, ETFs, insurance companies, pension funds, finance companies, securitisation vehicles}\}$.

Model specification:

$$\Delta \log(\text{Lev}_{s,t}) = \alpha_s + \beta_s \cdot \Delta \log(A_{s,t}) + \gamma \cdot X_{s,t-1} + \delta_t + \varepsilon_{s,t}$$

where:

- $\text{Lev}_{s,t} = A_{s,t}/E_{s,t}$ is the leverage ratio of sub-sector s
- $X_{s,t-1}$ = lagged controls: VIX level, term spread, credit spread, GDP growth
- α_s = **sub-sector fixed effects** (absorb time-invariant differences in business models, regulatory treatment, and average leverage levels)
- δ_t = **time fixed effects** (absorb common macroeconomic shocks — monetary policy, global risk appetite — that affect all sub-sectors simultaneously)
- β_s = sub-sector-specific procyclicality coefficient (the **key parameter of interest**)

Interpretation of β_s : $\beta_s > 0$ means leverage is procyclical (rises when assets grow, falls when assets shrink) — consistent with active VaR-constraint management. $\beta_s \approx 0$ means leverage is roughly constant (passive balance sheet). $\beta_s < 0$ would mean counter-cyclical leverage.

Expected ranking (H1): $\beta_{\text{HF}} > \beta_{\text{Lev. funds}} > \beta_{\text{Bond MF}} > \beta_{\text{MMF}} > \beta_{\text{IC}} \approx \beta_{\text{PF}} \approx 0$.

Estimation details:

- OLS panel regression with entity and time fixed effects.
- Standard errors clustered at the sub-sector level (9 clusters). Robustness: two-way clustering (sub-sector $\times$ year).
- In code: `sector_procyclicality()` in `src/models/var_amplification.py` estimates this for each sub-sector using rolling volatility as a VaR-implied leverage proxy: $\text{Lev}_{\text{implied}} = 1/(z_\alpha \cdot \sigma)$ where z_α is the VaR critical value and σ is rolling volatility.

Output: A table of β_s coefficients with standard errors, one row per sub-sector. Bar chart showing the ranking.

Specification 2: Interaction with stress **Purpose:** Test whether procyclicality *intensifies* during stress (asymmetric amplification).

Model specification:

$$\Delta \log(\text{Lev}_{s,t}) = \alpha_s + \beta_1 \cdot \Delta \log(A_{s,t}) + \beta_2 \cdot \Delta \log(A_{s,t}) \times \text{Stress}_t + \gamma \cdot X_{s,t-1} + \delta_t + \varepsilon_{s,t}$$

where Stress_t = dummy for quarters with VIX $\geq$ 75th percentile (or NBER recession indicator).

Key test: If $\beta_2 > 0$: procyclicality is *stronger* during stress — exactly when it does most damage.

Specification 3: Fire-sale simulation (Greenwood, Landier & Thesmar 2015) **Purpose:** Quantify how much NBFIs amplify fire-sale losses beyond a bank-only system.

Unit of observation: Simulated system (agent-based, not regression).

Input variables (simulation parameters):

Parameter	Symbol	Definition	Value
Initial shock	—	Exogenous decline in asset prices	-5%
Market depth	λ^{-1}	Kyle (1985) price impact parameter (\$ volume needed to move price 1%)	
<i>500K (calibrated)</i>			
VaR confidence	z_α	Confidence level for VaR constraint	99% (banks), 95–99.5% (NBFIs)
GARCH decay	—	EWMA lambda for volatility updating	0.94
Bank system	—	10 banks, leverage 10–15 $\times$, σ 1–2%	Calibrated to G-SIB data
NBFI system	—	20 entities: 5 HFs (leverage 5–25 $\times$), 8 investment funds (1–3 $\times$), 4 pension funds (1–2 $\times$), 3 insurance (3–8 $\times$)	Calibrated to FSB data

Simulation mechanism (multi-round):

1. **Initial shock:** Asset prices fall by -5%.
2. **Volatility update:** Each entity updates its rolling volatility estimate (EWMA, $\lambda = 0.94$).
3. **VaR tightening:** Higher volatility means the VaR constraint binds more — entities with VaR-constrained leverage ($\beta_s > 0$) must reduce exposure.

4. **Forced selling:** Entities sell assets to restore target leverage. Selling volume = excess leverage $\times$ assets.
5. **Price impact:** Aggregate selling depresses prices via Kyle lambda: $\Delta p = -\text{TotalSelling}/\text{MarketDepth}$.
6. **Repeat:** Steps 2–5 iterate until convergence (total selling $< 0.01\%$ of assets) or max 50 rounds.

Comparison:

- *Bank-only system:* Only 10 banks participate in fire-sale rounds.
- *Bank + NBFI system:* All 30 entities participate with their calibrated parameters.
- **Amplification ratio** = Total price drop (bank+NBFI) / Total price drop (bank-only).
If > 1 : NBFIs amplify.

Estimation details:

- Implemented as `FireSaleSimulation` dataclass in `src/models/var_amplification.py`.
- `run_counterfactual()` runs both systems with identical initial shock and compares.
- `amplification_ratio()` computes the ratio plus equity losses for each system.

Output:

1. **Amplification ratio:** e.g., $1.8\times$ means bank+NBFI system suffers 80% larger price drop.
2. **Round-by-round dynamics:** time series of asset prices, aggregate leverage, total equity, and number of active entities — for both systems.
3. **3-panel comparison chart:** asset price path, leverage path, and equity path for bank-only vs. bank+NBFI.

→ **Link to Module 2:** The procyclicality coefficients β_s from Specification 1 inform which sub-sectors are most dangerous in the fire-sale simulation. Module 2 (DCC-GARCH) tests whether the bank-NBFI correlations that drive fire-sale contagion are indeed time-varying and stress-dependent.

3.3.2 Module 2: DCC-GARCH Dynamic Correlations (Engle 2002)

Purpose: Estimate time-varying pairwise correlations between bank and NBFI sector returns, testing whether correlations spike during stress (confirming that contagion is regime-dependent, not constant).

Unit of observation: Pair of return series (bank i , NBFI sub-sector j) $\times$ week t .

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Bank returns	$r_{i,t}$	Weekly log-returns for individual G-SIBs or bank sector index	Bloomberg / Yahoo Finance	Weekly	Institution
NBFI returns	$r_{j,t}$	Weekly log-returns for NBFI sub-sector indices or representative firms	Bloomberg (BLK, BX, KKR, MET, PRU, etc.)	Weekly	Institution or sub-sector

Typical system: 6 entities — e.g., top 3 banks (JPM, GS, HSBC) + top 3 NBFIs (BLK, BX, MET). Or sector-level indices.

Model specification — Two-step DCC(1,1):

Step 1 — Univariate GARCH(1,1) for each series i :

$$r_{i,t} = \mu_i + \varepsilon_{i,t}, \quad \varepsilon_{i,t} = \sigma_{i,t} z_{i,t}, \quad z_{i,t} \sim N(0, 1)$$

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2$$

This gives: conditional volatilities $\sigma_{i,t}$ and standardised residuals $z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$.

Step 2 — DCC dynamics on the standardised residuals:

$$Q_t = (1 - a - b)\bar{Q} + a(z_{t-1}z'_{t-1}) + bQ_{t-1}$$

$$R_t = \text{diag}(Q_t)^{-1/2} \cdot Q_t \cdot \text{diag}(Q_t)^{-1/2}$$

where:

- $\bar{Q}$ = unconditional correlation matrix of z_t (estimated from full sample)
- a = "news" parameter — how fast correlations react to shocks (typically 0.01–0.10)
- b = "persistence" parameter — how slowly correlations revert (typically 0.85–0.98)
- R_t = time-varying correlation matrix (the object of interest)

Estimation details:

- **Step 1:** arch package in Python. `fit_univariate_garch()` in `src/models/dcc_garch.py`. GARCH(1,1) with constant mean. Standard MLE.
- **Step 2:** Quasi-MLE on DCC log-likelihood. `estimate_dcc()` uses Nelder-Mead optimisation with initial values $a_0 = 0.05$, $b_0 = 0.90$. Constraint: $a > 0$, $b > 0$, $a + b < 1$.
- Persistence $= a + b$. High persistence (e.g., 0.98) means correlations are very slowly mean-reverting.

Output:

1. **Dynamic pairwise correlations** $\rho_{ij,t}$: time series of correlation between every pair of entities. Plotted as line charts over 2000–2024.
2. **Sector-average correlations:** `sector_average_correlation()` averages all bank-HF pairs, all bank-insurance pairs, etc. Shows which bank-NBFI links are tightest.
3. **Contagion test** (Forbes & Rigobon 2002): `correlation_breakdown_test()` formally tests whether correlations are *significantly higher* during crisis periods ($VIX > 75$ th percentile) vs. calm periods. Output: mean correlation in each regime, difference, t-statistic, p-value.

Key findings expected:

- Bank-hedge fund correlations spike during stress (confirming fire-sale contagion channel from Module 1).
- Bank-pension fund correlations are low and stable (confirming that liability-driven entities don't amplify).
- Bank-insurance correlations increase moderately (mixed evidence).

→ **Link to Module 3:** DCC-GARCH captures *mean* correlation dynamics. Module 3 (Quantile VAR) goes further by estimating *tail-specific* dependence — testing whether the system is more connected at the 5th percentile than at the median.

3.3.3 Module 3: Quantile VAR — Nonlinear Transmission (Core Innovation)

Purpose: Estimate how monetary policy shocks propagate through the bank-NBFI system at different points of the return distribution. The key question: is MP transmission a *tail phenomenon* — much stronger at the 5th percentile than at the median?

Unit of observation: System of k return/macro series $\times$ quarter t (or month t).

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Policy rate	r_t^{MP}	Effective Fed funds rate (or shadow rate during ZLB)	FRED (DFF; Wu-Xia shadow rate)	Monthly → Quarterly	Aggregate
Bank sector return	r_t^B	Aggregate bank equity return index	Bloomberg (KBW Bank Index or constructed)	Weekly → Quarterly	Sector
Hedge fund return	r_t^{HF}	Hedge fund composite return	HFRX Global Index; Bloomberg	Monthly → Quarterly	Sector
MMF return	r_t^{MMF}	Money market fund return (proxy: short-term yield changes)	iMoneyNet; FRED (money market yields)	Monthly → Quarterly	Sector
Bond fund return	r_t^{BF}	Bond mutual fund return	Bloomberg (AGG total return)	Monthly → Quarterly	Sector
Insurance return	r_t^{IC}	Insurance sector equity return	Bloomberg (S&P Insurance Index)	Weekly → Quarterly	Sector
Credit spread	s_t	BAA–AAA corporate bond spread	FRED (BAA - AAA)	Monthly → Quarterly	Aggregate
Term spread	—	10Y - 2Y Treasury	FRED (T10Y2Y)	Monthly → Quarterly	Aggregate

System vector: $Y_t = (r_t^{MP}, r_t^B, r_t^{HF}, r_t^{MMF}, r_t^{BF}, r_t^{IC}, s_t)'$ — a **7-variable Quantile VAR**.

The choice of variables is driven by the economic question: we want to trace how a monetary policy shock (r_t^{MP}) propagates to each NBFIs sub-sector return and to credit spreads, and whether this transmission is stronger in the tails.

Model specification — Quantile VAR(p) at quantile τ :

$$Q_\tau(y_{i,t} | Y_{t-1}, \dots, Y_{t-p}) = c_i(\tau) + \sum_{j=1}^7 \sum_{l=1}^p \beta_{ij}^{(l)}(\tau) y_{j,t-l}$$

For each of the 7 equations, the dependent variable $y_{i,t}$ is regressed on p lags of all 7 variables. This gives $7 \times (p + 1)$ coefficients *at each quantile*.

What the QVAR captures (and what it does not) A standard VAR models the conditional **mean** of each variable: $E[y_{i,t} | Y_{t-1}]$. The coefficients are the same regardless of whether we are in calm or stressed times. This is a strong assumption — it says the transmission mechanism

is symmetric.

The Quantile VAR replaces the conditional mean with the conditional **quantile**: $Q_\tau(y_{i,t}|Y_{t-1})$. Now **every coefficient depends on τ** . Critically, this does *not* mean “where is the regressor in its distribution.” It means: **what predicts the τ -th quantile of the dependent variable?**

- At $\tau = 0.50$: you model the **median** of bank returns — “normal times” transmission.
- At $\tau = 0.05$: you model the **5th percentile** (left tail) of bank returns — “how do other variables affect the worst outcomes for banks?”

So $\beta_{ij}^{(l)}(\tau = 0.05)$ tells you how a one-unit change in y_j at lag l shifts the 5th percentile of y_i . This can be completely different from $\beta_{ij}^{(l)}(\tau = 0.50)$. For instance, a monetary policy tightening might have a small effect on the median of hedge fund returns but a devastating effect on the 5th percentile — because only the worst-performing funds face margin calls and forced liquidation.

The Ando, Greenwood-Nimmo & Shin (2022) framework Once the QVAR is estimated at quantile τ , the companion matrix gives the $MA(\infty)$ representation, from which we compute a Generalised Forecast Error Variance Decomposition (GFEVD) — **at each quantile separately**. This is the key contribution of Ando et al. (2022): applying the Diebold-Yilmaz (2012) connectedness framework not at the mean but across the quantile distribution.

The GFEVD entry $\theta_{ij}(\tau)$ answers: “What share of variable i ’s h -step-ahead forecast error variance (at quantile τ) is attributable to shocks originating in variable j ?” From this $k \times k$ matrix $\Theta(\tau)$, Ando et al. define:

- **Total connectedness** $C(\tau)$: the average off-diagonal share — how much of each variable’s risk comes from *other* variables.
- **Directional measures**: TO (how much i transmits), FROM (how much i receives), NET (transmitter vs. receiver).

The critical insight: **you compute $C(\tau)$ at every quantile and compare**. If $C(0.05) \gg C(0.50)$, it means the system is dramatically more interconnected when variables are in their left tails. In normal times, shocks to one sector explain perhaps 30% of variation in others. In stress, they might explain 70%. This asymmetry is invisible to a standard linear VAR — which gives you one number regardless of whether markets are calm or crashing.

State-dependence and distance to constraints An important refinement: shock propagation depends not only on structural integration (who is connected to whom) but also on **how close intermediaries are to their VaR/margin/redemption constraints** when the shock hits. A moderate monetary policy tightening in a world where hedge funds have plenty of excess margin capacity is very different from the same shock when funds are already near their VaR limits.

The QVAR captures this implicitly: the coefficients $\beta_{ij}(\tau)$ vary with τ precisely because the

transmission mechanism changes when the system is stressed. But we can make the state-dependence explicit with a **threshold specification**:

$$Q_\tau(y_{i,t}) = \begin{cases} \alpha_1(\tau) + \beta_1(\tau)X_t & \text{if Slack}_t > \bar{s} \text{ (far from constraints)} \\ \alpha_2(\tau) + \beta_2(\tau)X_t & \text{if Slack}_t \leq \bar{s} \text{ (near constraints)} \end{cases}$$

where Slack_t measures the distance to binding constraints (e.g., aggregate excess margin, or the gap between actual VaR and VaR limit). The hypothesis: $\beta_2(\tau = 0.05) \gg \beta_1(\tau = 0.05)$ — even in the left tail, amplification is much worse when constraints are nearly binding. This is a **double nonlinearity**: tail-specific *and* state-dependent.

This connects to the HLI from Project 1: high $\lambda_1(W_t)$ (tight network) **plus** low slack (near constraints) is the dangerous combination. Neither alone is sufficient.

Estimation details:

- **Equation-by-equation quantile regression**: Each equation estimated separately via `statsmodels.QuantReg`. This is computationally feasible even for 7 equations because each is a standard linear quantile regression.
- **Lags**: $p = 4$ for quarterly data (one year of history). BIC-optimal at each quantile; robustness with $p \in \{1, 2, 4\}$.
- **Quantiles**: $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$.
- **Residuals**: After estimation, compute residuals $\hat{u}_{i,t}(\tau) = y_{i,t} - \hat{Q}_\tau(y_{i,t}|\cdot)$ and their covariance matrix $\hat{\Sigma}(\tau)$.
- Implemented in `estimate_quantile_var()` and `estimate_quantile_var_grid()` in `src/econometrics`.

From the QVAR, three outputs are computed:

(a) Quantile Impulse Response Functions (QIRFs):

- Trace the effect of a 1-standard-deviation shock to variable j (e.g., policy rate) on variable i (e.g., hedge fund return) over h periods.
- Uses the companion matrix representation: shock is propagated through the $\text{MA}(\infty)$ representation.
- Computed via `quantile_irf()`. Key comparison: `compare_quantile_irfs()` shows the IRF at $\tau = 0.05$ vs. $\tau = 0.50$.
- **Expected result**: A monetary policy tightening shock has a 2–3 $\times$ larger negative effect on hedge fund returns at $\tau = 0.05$ than at $\tau = 0.50$.

(b) Quantile Connectedness (Ando, Greenwood-Nimmo & Shin 2022):

- From the QVAR, compute the GFEVD $\Theta(\tau)$ at horizon $h = 10$.

- Total connectedness: $C(\tau) = \frac{1}{k} \sum_{i \neq j} \theta_{ij}(\tau) \times 100\%$.
- Directional: TO, FROM, NET for each variable.
- Computed via `quantile_connectedness()` in `src/econometrics/quantile_var.py`.
- **Expected result:** $C(0.05) \gg C(0.50)$ — the system is much more interconnected in the left tail.

(c) Rolling Quantile Connectedness:

- `rolling_quantile_connectedness(window=60, step=1)` produces a time series $C_t(\tau)$ at each quantile.
- Tracks how tail connectedness evolves over 2000–2024, spiking around crisis episodes.

Output summary:

Output	What it shows	Format
QIRF comparison plot	MP shock $\rightarrow$ HF return at $\tau = 0.05$ vs $\$0.50\$$	Line chart (horizon on x-axis)
Connectedness table	$C(\tau)$ at each quantile	Single row: $\$C(0.05), C(0.25), C(0.50), C(0.75), C(0.95)\$$
GFEVD heatmap	$\Theta(\tau)$ matrix at $\tau = 0.05$ vs $\$0.50\$$	7×7 heatmap (two panels)
Directional bar chart	TO, FROM, NET by variable at each quantile	Grouped bar chart
Rolling connectedness	$C_t(0.05)$ and $C_t(0.50)$ over time	Time series (2 lines)

$\rightarrow$ **Link to Module 4:** Module 3 shows that transmission is stronger in the tails. Module 4 tests whether NBFI leverage explains *why* — via the variance (scale) channel, not just the mean (location) channel.

3.3.4 Module 4: Location-Scale Growth-at-Risk

Purpose: Test whether NBFI leverage increases the *conditional variance* (not just the conditional mean) of future growth outcomes, generating fatter left tails. This is the “variance amplification” hypothesis (H3).

Unit of observation: Quarter $\$t\$$ (aggregate time series) or country $\$c\$ \times$ quarter $\$t\$$ (panel).

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Future GDP growth	y_{t+h}	Real GDP growth at horizon h quarters ahead	FRED (GDPC1)	Quarterly	Aggregate
Financial conditions	FCI_t	Chicago Fed NFCI or composite index	FRED (NFCI)	Quarterly	Aggregate
NBFI leverage	NBFI_t	Aggregate NBFI leverage (from P1 Module 1), or NBFI assets as % of GDP (FSB)	Module 1 output; FSB	Quarterly / Annual	Aggregate
Interaction	$\text{FCI}_t \times \text{NBFI}_t$	Tests whether the FCI effect on variance depends on NBFI size	Constructed	Quarterly	Aggregate

Model specification — Two-step Location-Scale model:

Step 1 — Location (conditional mean):

$$\hat{\mu}_{t+h} = E[y_{t+h} \mid X_t] = \alpha + \beta_1 \cdot \text{FCI}_t + \beta_2 \cdot \text{NBFI}_t + \beta_3 \cdot (\text{FCI}_t \times \text{NBFI}_t)$$

Estimated by OLS. Residuals: $\hat{e}_{t+h} = y_{t+h} - \hat{\mu}_{t+h}$.

Step 2 — Scale (conditional variance):

$$\log(\hat{e}_{t+h}^2) = \delta_0 + \delta_1 \cdot \text{FCI}_t + \delta_2 \cdot \text{NBFI}_t + \delta_3 \cdot (\text{FCI}_t \times \text{NBFI}_t) + v_t$$

Estimated by OLS on the log-squared residuals from Step 1. This gives $\hat{\sigma}_{t+h}^2 = \exp(\hat{\delta}_0 + \hat{\delta}_1 \text{FCI}_t + \dots)$.

Conditional quantiles are then:

$$\hat{Q}_\tau(y_{t+h} \mid X_t) = \hat{\mu}_{t+h} + \hat{\sigma}_{t+h} \cdot \Phi^{-1}(\tau)$$

where $\Phi^{-1}(\tau)$ is the standard normal quantile function (or Student-t quantile for heavier tails).

Key parameters:

- $\delta_2 > 0$: NBFI leverage *increases conditional variance* of future growth (fatter tails in both directions).

- $\delta_3 > 0$: NBFI leverage amplifies the variance-increasing effect of tight financial conditions.
- Standard GaR (Adrian et al. 2019) only models the location; our LS extension separates *mean shift* from *variance amplification*.

Estimation details:

- Implemented in `location_scale_gar()` in `src/econometrics/location_scale.py`.
- Step 1: OLS with HC1 standard errors.
- Step 2: OLS on log-squared residuals.
- Alternative: `skewt_location_scale_fit()` estimates both location and scale parameters jointly via MLE assuming a Hansen (1994) skewed-t distribution (captures asymmetry).
- Horizons: $h \in \{1, 4, 8, 12\}$ quarters.
- Quantiles computed: $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$.

Output:

1. **Location coefficients** ($\beta_1, \beta_2, \beta_3$): do FCI and NBFI shift the mean of future growth?
2. **Scale coefficients** ($\delta_1, \delta_2, \delta_3$): do FCI and NBFI increase the *variance* of future growth?
3. **Fan chart**: predicted distribution of y_{t+h} for different values of NBFI_t — showing that high NBFI leverage produces a *wider* (fatter-tailed) distribution, not just a shifted one.
4. **Tail risk amplification test** (`tail_risk_amplification()`): quantile regression with $\text{Stress} \times \text{NBFI}$ interaction at $\tau = 0.05$, confirming that the amplification is asymmetric.

Key contribution: This separates two channels: (a) NBFI leverage *shifts the mean* of future growth (location effect — already in Adrian et al. 2019); (b) NBFI leverage *fattens the tails* of future growth (scale effect — **new contribution**). If $\delta_2 > 0$ but $\beta_2 \approx 0$: NBFI leverage doesn't change average growth but makes extreme outcomes more likely.

→ **Link to Module 5:** Module 4 shows NBFI leverage amplifies risk. Module 5 asks: why didn't this blow up in 2022–23? The answer lies in the network *topology* — within-OFI stress was high but didn't propagate to the real economy because banks acted as a firewall.

3.3.5 Module 5: Resolving the Banerjee et al. (2025) Puzzle

Purpose: Explain why the 2022–23 tightening cycle did *not* produce a financial crisis despite high NBFI stress. Using Diebold-Yilmaz sector-level decomposition, show that within-OFI connectedness was high but OFI-to-real-economy connectedness remained low.

Unit of observation: System of sector-level return series $\times$ week \$t\$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
OFI sector returns	$r_t^{OFI,j}$	Returns for each OFI sub-sector j (HFs, bond funds, MMFs, etc.)	Bloomberg / HFRX	Weekly	Sub-sector
Bank sector return	r_t^B	Bank sector return	Bloomberg (KBW Bank Index)	Weekly	Sector
Real economy proxy	r_t^{RE}	Industrial production growth, or credit growth, or non-financial corporate bond return	FRED / Bloomberg	Monthly → Weekly	Aggregate
Policy rate changes	Δr_t^{MP}	Change in effective Fed funds rate	FRED	Monthly	Aggregate

System vector: $Y_t = (r_t^{HF}, r_t^{BondFund}, r_t^{MMF}, r_t^{IC}, r_t^B, r_t^{RE})'$ — a **6-variable VAR** capturing within-OFI, OFI-to-bank, and OFI-to-real-economy links.

Model specification — Standard VAR(p) with Diebold-Yilmaz GFEVD:

$$Y_t = c + A_1 Y_{t-1} + \dots + A_p Y_{t-p} + u_t$$

From the VAR, compute the Generalised FEVD Θ at horizon h , then decompose total connectedness into:

- **Within-OFI connectedness:** $C^{OFI} = \frac{1}{n_{OFI}} \sum_{i,j \in OFI, i \neq j} \theta_{ij}$ — spillovers *among* OFI sub-sectors.
- **OFI-to-bank connectedness:** $C^{OFI \rightarrow B} = \sum_{i \in OFI} \theta_{B,i}$ — spillovers *from* OFIs *to* banks.
- **OFI-to-real economy connectedness:** $C^{OFI \rightarrow RE} = \sum_{i \in OFI} \theta_{RE,i}$ — spillovers *from* OFIs *to* the real economy.

Estimation details:

- Standard VAR estimated by OLS (not quantile VAR — this module uses mean connectedness, not tail connectedness).
- `compute_connectedness()` in `src/analysis/systemic_risk.py` with lags $= 4$, horizon $h = 10$.

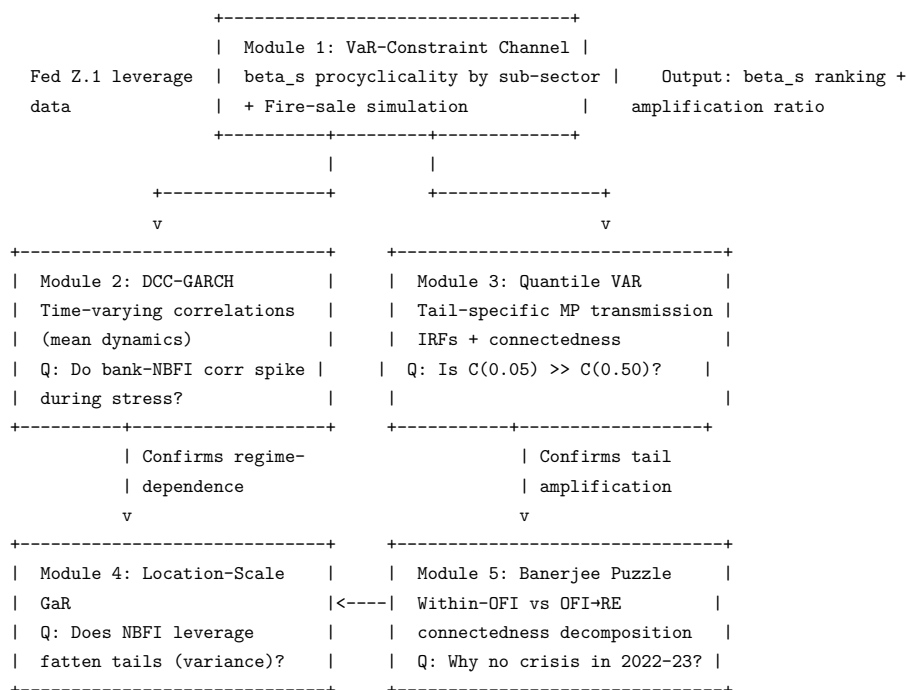
- `aggregate_connectedness_by_sector()` aggregates the Θ matrix to sector blocks.
- Rolling estimation (window = 60 weeks) to track how within-OFI vs. OFI-to-RE connect-
edness evolves over 2020–2024.

Output:

1. **Sector-block GFEVD heatmap:** shows within-OFI block is "hot" (high spillovers) while OFI→RE block is "cold" (low spillovers) during 2022–23.
2. **Rolling decomposition:** time series of C_t^{OFI} , $C_t^{OFI \rightarrow B}$, and $C_t^{OFI \rightarrow RE}$. Expected: within-OFI spikes in 2022 but OFI→RE stays flat.
3. **Narrative:** Post-Basel III bank capital buffers acted as a *firewall*, absorbing OFI stress without transmitting it to the real economy. This is consistent with bank capital ratios remaining above regulatory minima throughout the tightening cycle.

Key insight: The puzzle resolves because contagion has two stages: (1) within-NBFI stress propagation (high in 2022–23), and (2) NBFI-to-real-economy transmission (low in 2022–23 due to bank resilience). Standard measures that aggregate both stages miss this.

3.3.6 Module Linkage Map (Project 2)



3.4 Data Sources & Sample

Data Source	Variables	Frequency	Coverage
Flow of Funds (Fed Z.1)	Sector-level as- sets, liabilities, leverage	Quarterly	1980–present
ECB Invest- ment Fund Statistics	EU fund AUM, flows, leverage by type	Quarterly	2009–present
EPFR Global	Fund flows by type, country, asset class	Weekly/Monthly	2000–present
Bloomberg / Refinitiv	Sector return indices (HF, MMF, insur- ance, pension)	Daily	2000–present
BIS Credit Statistics	Credit to pri- vate sector, spreads	Quarterly	1999–present
FRED	Fed funds rate, term spreads, GDP, CPI	Mixed	1960–present
Banerjee et al. (2025)	Replication data for BIS Bulletin 116	Quarterly	2000–2024

Sample period: 2000Q1–2024Q4 for most modules. DCC-GARCH uses daily data (2000–2024). Quantile VAR uses quarterly data.

Sub-sector classification: Hedge funds (HFs), money market funds (MMFs), bond mutual funds, equity mutual funds, ETFs, insurance companies (ICs), pension funds (PFs), finance companies, securitisation vehicles.

3.5 Literature Positioning

1. **Monetary policy & financial intermediaries:** Extends Adrian & Shin (2010, 2014) VaR-constraint channel from banks to NBFIs. Tests whether the leverage-procyclicality mechanism operates more strongly in specific NBFI sub-sectors.
2. **NBFI amplification:** Directly builds on Banerjee, Hofmann, Ng & Pinter (2025) aggregate finding. Our sub-sector decomposition and quantile methods address their acknowledged limitation of wide confidence bands.
3. **Quantile VAR:** Adapts Ando, Greenwood-Nimmo & Shin (2022) from cross-country to cross-sector application. First use of quantile connectedness for monetary policy transmission analysis.
4. **Location-scale models:** Extends Adrian, Boyarchenko & Giannone (2019) GaR framework by modelling the scale (variance) channel separately, testing whether NBFI leverage

fattens both tails.

3.6 Identification & Robustness

- **Monetary policy shock identification:** High-frequency identification (Gurkaynak, Sack & Swanson 2005) using Fed funds futures around FOMC announcements. Robustness: Romer & Romer (2004) narrative shocks, Jarocinski & Karadi (2020) sign-restricted shocks.
- **Endogeneity of NBFI leverage:** NBFI leverage is instrumented with lagged regulatory capital ratios of connected banks (supply-side shifters).
- **Quantile VAR lag selection:** BIC-optimal lag length at each quantile; robustness across $p \in \{1, 2, 4\}$.
- **Small-sample inference:** Bootstrap confidence intervals (1000 replications) for quantile IRFs and connectedness measures.

3.7 Expected Contributions

1. Sub-sector decomposition: hedge funds drive amplification, pension funds dampen it.
2. Quantile connectedness reveals tail-specific MP amplification invisible to linear methods.
3. Location-scale evidence: NBFI leverage fattens both tails (variance effect), not just shifts the mean.
4. Resolution of the 2022–23 resilience puzzle via network topology analysis.

4 Project 3: Contagion Across Borders

Subtitle: How NBFI Stress in One Jurisdiction Spills Over via the Global Financial Cycle

4.1 Motivation

Cross-border NBFI linkages create a transmission mechanism for financial stress through three interconnected channels: (1) dollar funding, where NBFIs borrow in USD via repo and FX swaps; (2) portfolio rebalancing by global investment funds whose procyclical flows amplify local shocks; and (3) the global financial cycle (Rey 2013), a common factor in risky asset prices that NBFIs both respond to and amplify.

4.2 Research Questions

RQ1. Is cross-border NBFI connectedness asymmetric — dramatically higher in the left tail (stress) than at the median (normal times)?

RQ2. Does dollar funding stress transmit cross-border, and is this transmission nonlinear (stronger in the tails)?

RQ3. Are NBFIs portfolio flows more volatile and more cross-border-connected than bank flows?

RQ4. Does NBFIs penetration amplify the transmission of the global financial cycle to local financial conditions? Is this robust to IV estimation?

RQ5. Can a hedge fund failure cascade through prime brokerage links to generate cross-border deleveraging?

4.3 Data Sources & Sample

Data Source	Variables	Frequency	Coverage
BIS Locational Banking Stats	Cross-border claims by sector & country	Quarterly	2000–present
EPFR Global	Cross-border fund flows by type & destination	Monthly	2005–present
IMF CPIS	Portfolio investment positions by country pair	Annual	2001–present
BIS OTC Derivatives	FX swap/-forward outstanding by currency	Semiannual	2004–present
Bloomberg	CIP basis (cross-currency basis swaps), VIX, equity indices	Daily	2000–present
Miranda-Agrippino & Rey (2020)	Global Financial Cycle factor (updated)	Monthly	1990–present
FSB Global Monitoring	NBFI penetration ratio by country	Annual	2002–present

Sample period: 2005Q1–2024Q4 for cross-country quantile connectedness (EPFR flow data availability). Extended to 2000 for GFC panel regressions using BIS data.

Country coverage: G20 + selected financial centres (US, UK, EA, JP, CH, AU, CA, KR, SG, HK, BR, MX, IN, ZA, TR, RU, CN). Separate EME sub-sample for GFC amplification tests.

4.4 Literature Positioning

1. **Global financial cycle:** Extends Rey (2013) and Miranda-Agrippino & Rey (2020) by testing whether NBFi penetration amplifies GFC transmission. First causal evidence using IV (US monetary policy shocks $\rightarrow$ GFC $\rightarrow$ local conditions, with NBFi interaction).
2. **Cross-border contagion:** Complements Forbes & Warnock (2012) on capital flow surges/stops and Broner, Didier, Erce & Schmukler (2013) on gross flows. Adds NBFi-specific decomposition and quantile methods.
3. **Dollar funding:** Builds on Avdjiev, Du, Koch & Shin (2019) and Eguren-Martin, Ossandon Busch & Reinhardt (2024) on dollar funding channels. Tests nonlinear transmission via quantile regression.
4. **Network contagion:** Agent-based simulation follows Eisenberg & Noe (2001) clearing mechanism with Kyle (1985) price impact, extending Cont & Schaanning (2017) fire-sale framework to cross-border setting.

4.5 Identification & Robustness

- **IV for GFC factor:** US monetary policy shocks (Jarocinski & Karadi 2020) instrument the GFC, addressing reverse causality from local conditions to global factor.
- **NBFi penetration endogeneity:** NBFi share instrumented with (a) legal origin (La Porta et al. 1998) for cross-section, (b) lagged pension reform dummies for within-country variation.
- **Alternative connectedness measures:** Barunik & Krehlik (2018) frequency-domain connectedness as robustness check on time-domain results.
- **EME vs AE heterogeneity:** Full interaction models allowing different coefficients for advanced and emerging economies.

4.6 Empirical Strategy

Project 3 traces three contagion channels, then unifies them in a quantile connectedness framework and a cascade simulation.

Channel 1 (Dollar Funding) $-+$

Channel 2 (Portfolio Flows) $-++\rightarrow$ Module 4 (Cross-Country QVAR) $---\rightarrow$ Module 5 (GFC Amplification Panel) $---\rightarrow$ Module 6 (Cascade Sim)

Channel 3 (Global Fin Cycle) $-+$

4.6.1 Module 1: Dollar Funding Channel

Purpose: Test whether dollar funding stress (measured by the CIP basis) is amplified in countries whose financial sectors have larger structural dollar hedging needs, and whether this amplification is nonlinear (stronger in the tails).

Economic Logic The CIP basis is determined by the balance between supply and demand for dollar hedging in the FX swap market. What matters is not generic "NBFIs size" but specifically **how much of country $\$c\$$'s financial sector relies on FX-hedged dollar assets or dollar funding via swaps**:

- **Pension funds and insurance companies** in countries like Japan, the Netherlands, and Korea hold large portfolios of US Treasuries and corporate bonds, hedged via 3-month FX forwards that must be rolled quarterly. This creates structural, recurring demand for dollar hedging.
- **NBFI dollar borrowing**: Institutions that use FX swaps as synthetic dollar funding (borrow dollars, lend local currency) are directly exposed when the swap market seizes up.

When stress hits (VIX spikes, equity markets fall), these institutions face margin calls on their FX forwards and compete for scarce dollar liquidity. Countries with larger structural dollar hedging demand see their CIP basis blow out more — not because NBFIs are "big" but because their hedging activity creates concentrated demand that overwhelms dealer capacity during stress.

Unit of observation: Currency pair $\$c\$ \times$ month $\$t\$$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
CIP basis	$CIP_{c,t}$	Cross-currency basis swap spread for currency $\$c\$$ vs USD (3-month tenor). Negative = dollar funding premium	Bloomberg	Daily $\rightarrow$ Monthly avg	Currency pair
FX hedge demand	$FXHedge_{c,t}$	Structural dollar hedging need of country $\$c\$$'s financial sector (see below)	BIS Triennial Survey; IMF CPIS; national sources	Semi-annual / Annual	Country
VIX	VIX_t	CBOE Volatility Index	FRED	Daily $\rightarrow$ Monthly avg	Global
USD broad index	—	Trade-weighted USD index	FRED (DTWEXBGS)	Daily $\rightarrow$ Monthly avg	Global
Local equity return	$r_{c,t}^{EQ}$	Country $\$c\$$ equity index return	Bloomberg	Daily $\rightarrow$ Monthly	Country

Measuring $FXHedge_{c,t}$ — three alternatives (from most to least precise):

1. **Outstanding FX swap/forward positions** of country \$c\$’s non-bank financial sector (from BIS Triennial Survey — available for major currencies, semi-annual)
2. **Cross-border bond holdings that need hedging:** country \$c\$’s institutional investors’ holdings of foreign-currency bonds (from IMF Coordinated Portfolio Investment Survey), scaled by estimated hedge ratios
3. **Fallback:** NBFI foreign assets as % of GDP from FSB Global Monitoring Report (coarser but broader country coverage)

Model specification — Quantile regression:

$$Q_\tau(\text{CIP}_{c,t}) = \alpha_c + \beta_1(\tau) \cdot \text{VIX}_t + \beta_2(\tau) \cdot r_{c,t}^{EQ} + \beta_3(\tau) \cdot \text{FXHedge}_{c,t} + \beta_4(\tau) \cdot (\text{VIX}_t \times \text{FXHedge}_{c,t}) + \varepsilon_{c,t}$$

Key test: $\beta_4(\tau = 0.05) \neq 0$ but $\beta_4(\tau = 0.50) \approx 0$. Countries with larger structural dollar hedging needs suffer larger CIP blowouts during stress — but not during normal times. The mechanism: stress triggers a rush to roll hedges, demand overwhelms dealer capacity, and the basis blows out in proportion to pre-existing hedging demand.

Estimation: `statsmodels.QuantReg` at $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$, with country fixed effects. Standard errors clustered by country.

Output: Coefficient table across quantiles; scatter plot of CIP basis vs. VIX for high- vs. low-FXHedge countries.

4.6.2 Module 2: Portfolio Flow Channel

Purpose: Compare the volatility and cross-border connectedness of NBFI portfolio flows vs. bank flows.

Unit of observation: Country \$c\$ × month \$t\$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
NBFI portfolio flows	$F_{c,t}^{NBFI}$	Net cross-border fund flows (equity + bond funds) to country \$c\$	EPFR Global	Monthly	Country

Variable	Symbol	Definition	Source	Frequency	Level
Bank flows	$F_{c,t}^{Bank}$	Cross-border bank claims on country \$c\$ (change)	BIS Locational Banking Stats	Quarterly → Monthly (interp)	Country
VIX	—	Stress proxy	FRED	Monthly	Global
GFC factor	GFC_t	Global financial cycle (from Module 3 below)	PCA extraction	Monthly	Global

Analysis 1 — Volatility comparison: Compute rolling 12-month standard deviation of $F_{c,t}^{NBFI}$ vs. $F_{c,t}^{Bank}$ for each country. Test: $\sigma(F^{NBFI}) > \sigma(F^{Bank})$?

Analysis 2 — Diebold-Yilmaz connectedness on cross-country flows:

System: $Y_t = (F_{US,t}^{NBFI}, F_{UK,t}^{NBFI}, F_{EA,t}^{NBFI}, F_{JP,t}^{NBFI}, F_{EM,t}^{NBFI})'$ — a 5-variable VAR of NBFI flows across major regions.

Estimate VAR(4) and compute GFEVD at $h = 10$. Then repeat with bank flows. Compare: is total connectedness of NBFI flows higher than bank flows?

Estimation: OLS VAR with `compute_connectedness()` from `src/analysis/systemic_risk.py`. Rolling window (60 months) for time-varying comparison.

Output:

1. Bar chart: average flow volatility for NBFI vs. bank flows, by country group.
2. Total connectedness comparison: C^{NBFI} vs. C^{Bank} — expected: NBFI flows are more connected.
3. Rolling connectedness plot: NBFI flow connectedness spiking during 2008, 2013, 2020.

4.6.3 Module 3: Global Financial Cycle (GFC) Factor Extraction

Purpose: Extract the common factor driving international risky asset prices (Miranda-Agrippino & Rey 2020) and test whether NBFI penetration amplifies its transmission to local financial conditions.

Unit of observation: Global factor $\times$ month t (extraction); Country c $\times$ quarter t (amplification test).

Input variables for factor extraction:

Variable	Symbol	Definition	Source	Frequency	Level
International equity returns	$r_{c,t}$	Monthly equity index returns for N countries	Bloomberg / MSCI country indices	Monthly	Country

Countries: US, UK, Germany, France, Japan, Canada, Australia, Korea, Singapore, Hong Kong, Brazil, Mexico, India, South Africa, Turkey + possibly China, Russia (20+ countries).

Factor extraction:

1. Standardise each country's monthly equity returns: $\tilde{r}_{c,t} = (r_{c,t} - \bar{r}_c) / \sigma_c$.
2. Apply PCA to the $N \times T$ matrix of standardised returns.
3. The first principal component $\text{GFC}_t = \text{PC}_1(t)$ is the global financial cycle factor.

Estimation details:

- `extract_global_factor()` in `src/analysis/global_financial_cycle.py`.
- Uses `sklearn.decomposition.PCA` after `StandardScaler`.
- Typically PC_1 explains 30–40% of total variance across countries.
- Rolling version: `rolling_global_factor(window=252, step=21)` re-estimates PCA on rolling windows.

Output: Monthly time series $\{\text{GFC}_t\}$ — a single global factor. Plus: loadings per country (which countries load most heavily on the GFC) and explained variance ratio.

Extension: The Global Tail Cycle The standard GFC captures co-movement at the **mean** — it answers "do returns move together on average?" But the relevant question for systemic risk is: **do they crash together?** A market can look diversified in normal times but exhibit near-perfect correlation during stress. The GFC misses this.

A **Global Tail Cycle (GTC)** extracts the common factor from **tail co-movements** — correlations specifically in the left tail. There are two approaches:

(a) Tail dependence factor: For each country pair (n,m) , compute the time-varying lower-tail dependence coefficient $\lambda_{nm,t}^L$ (using DCC-based tail dependence or exceedance correlations on a rolling window: joint probability of both being below the 5th percentile). Then extract the first principal component of the λ^L matrix across all pairs — this is the Global Tail Cycle.

(b) Quantile factor model (Ando & Bai 2020): Estimate a factor model directly at $\tau = 0.05$, extracting the first factor from a cross-country panel of equity returns at the 5th percentile. This literally is "the common factor in left-tail outcomes."

The GTC is conceptually distinct from the standard GFC. The GFC could be high (assets correlated on average) while tail dependence is low (they don't crash together). Or the GFC

could be moderate but the GTC very high (normal times look diversified, but stress is perfectly correlated). The GTC captures exactly the moment when diversification fails — when VaR constraints bind simultaneously across jurisdictions and coordinated deleveraging amplifies the initial shock.

In Module 5, both GFC and GTC enter the panel regression to test whether NBF1 penetration amplifies *average* co-movement (GFC interaction) vs. *tail* co-movement (GTC interaction) differently.

4.6.4 Module 4: Cross-Country Quantile Connectedness

Purpose: Test whether cross-border financial connectedness is asymmetric — dramatically higher in the left tail than at the median.

Unit of observation: System of \$k\$ country returns $\times$ week \$t\$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Country equity returns	$r_{c,t}$	Weekly log-returns for major equity indices	Bloomberg	Weekly	Country
USD funding proxy	USD_t	Change in trade-weighted USD index or 3M CIP basis	Bloomberg / FRED	Weekly	Global

System: $Y_t = (r_t^{US}, r_t^{UK}, r_t^{EU}, r_t^{JP}, r_t^{EM}, USD_t)'$ — a **6-variable Quantile VAR** of cross-country returns plus dollar funding.

Model specification — identical to P2 Module 3 (QVAR):

$$Q_\tau(y_{i,t} | Y_{t-1}, \dots, Y_{t-p}) = c_i(\tau) + \sum_{j=1}^6 \sum_{l=1}^p \beta_{ij}^{(l)}(\tau) y_{j,t-l}$$

Estimation: Equation-by-equation **QuantReg**, lags \$p = 4\$, quantiles $\tau \in \{0.05, 0.50, 0.95\}$, GFEVD at \$h = 10\$.

Key prediction: Total connectedness at $\tau = 0.05 \gg \tau = 0.50$ — cross-border contagion is much stronger during stress.

Rolling version: Window = 60 weeks, step = 1. Time series of $C_t(0.05)$ and $C_t(0.50)$ through GFC 2008, Taper Tantrum 2013, COVID 2020.

Output:

1. Total connectedness at each quantile: $\$C(0.05)\$, \$C(0.50)\$, \$C(0.95)\$$.
2. Pairwise GFEVD matrices at $\tau = 0.05$ vs. $\$0.50\$$ (heatmaps).
3. Directional spillovers: which countries are net transmitters of tail risk.
4. Rolling plot: $C_t(0.05)$ and $C_t(0.50)$ over time.

4.6.5 Module 5: GFC Amplification Panel Regression

Purpose: Test whether NBFI penetration amplifies the transmission of the Global Financial Cycle to local financial conditions. This is the causal core of Project 3.

Unit of observation: Country $\$c\$$ $\times$ quarter $\$t\$$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Local credit growth	$y_{c,t}$	Credit-to-GDP growth or local equity return	BIS Credit Statistics; Bloomberg	Quarterly	Country
GFC factor	GFC_t	From Module 3	Module 3 output	Monthly $\rightarrow$ Quarterly	Global
NBFI penetration	$NBFI_{c,t}$	NBFI assets as % of GDP in country $\$c\$$	FSB Global Monitoring Report	Annual (interp. to quarterly)	Country
Interaction	$GFC_t \times NBFI_{c,t}$	Tests amplification	Constructed	Quarterly	Country
Bank-NBFI connectedness	$Conn_{c,t}$	Bilateral bank-NBFI exposure (from BIS) or rolling Granger-causality measure	BIS / constructed	Quarterly	Country

Model specification — OLS panel (baseline):

$$y_{c,t} = \alpha_c + \beta_1 \cdot GFC_t + \beta_2 \cdot NBFI_{c,t} + \beta_3 \cdot (GFC_t \times NBFI_{c,t}) + \gamma \cdot X_{c,t} + \varepsilon_{c,t}$$

where α_c = country fixed effects.

Key parameter: β_3 . If $\beta_3 > 0$: countries with larger NBFI sectors experience *stronger* transmission of the global financial cycle to local credit conditions. This is the amplification effect.

NBFI Counterparty Decomposition A single $\text{NBFI}_{c,t}$ measure (assets as % of GDP) is too coarse — it conflates NBFI activities with completely different systemic implications. The results differ fundamentally depending on whether NBFIs:

- Lend to domestic banks vs. borrow from domestic banks
- Lend to foreigners vs. borrow from foreigners
- Lend to or borrow from the domestic private sector

A country where NBFIs primarily borrow from domestic banks and lend to the private sector (credit intermediation model) has a different GFC transmission profile than one where NBFIs borrow from foreigners and trade derivatives with domestic banks (market intermediation model).

Extended specification — replace aggregate NBFI with counterparty-decomposed measures:

$$y_{c,t} = \alpha_c + \beta_1 \cdot \text{GFC}_t + \sum_k \beta_{2,k} \cdot \text{NBFI}_{c,t}^{(k)} + \sum_k \beta_{3,k} \cdot (\text{GFC}_t \times \text{NBFI}_{c,t}^{(k)}) + \gamma X_{c,t} + \varepsilon_{c,t}$$

where k indexes NBFI activity types:

- $\text{NBFI}^{\text{bank-borrow}}$: NBFI borrowing from domestic banks (repo, credit lines) → captures credit risk channel
- $\text{NBFI}^{\text{bank-lend}}$: NBFI lending to domestic banks (deposits, bank bonds) → captures funding risk channel
- $\text{NBFI}^{\text{foreign-borrow}}$: NBFI cross-border borrowing → captures imported funding shock channel
- $\text{NBFI}^{\text{foreign-lend}}$: NBFI cross-border lending → captures outward transmission channel
- $\text{NBFI}^{\text{private-lend}}$: NBFI lending to private sector → captures real economy credit channel

The $\beta_{3,k}$ coefficients reveal *which type* of NBFI activity amplifies GFC transmission most. The hypothesis: $\beta_{3,\text{foreign-borrow}}$ is largest (countries whose NBFIs rely on foreign funding are most vulnerable to global shocks), while $\beta_{3,\text{private-lend}}$ may actually be *stabilising* (credit intermediation provides a buffer).

Data source: "From-whom-to-whom" financial accounts — Fed Z.1, ECB Financial Accounts, OECD institutional sector accounts.

Global Tail Cycle extension Replace the standard GFC with the Global Tail Cycle (from Module 3 extension):

$$y_{c,t} = \alpha_c + \beta_1^{\text{GFC}} \cdot \text{GFC}_t + \beta_1^{\text{GTC}} \cdot \text{GTC}_t + \beta_3^{\text{GFC}} \cdot (\text{GFC}_t \times \text{NBFI}_{c,t}) + \beta_3^{\text{GTC}} \cdot (\text{GTC}_t \times \text{NBFI}_{c,t}) + \dots$$

If $\beta_3^{\text{GTC}} \gg \beta_3^{\text{GFC}}$: NBFi penetration amplifies *tail* co-movement far more than *average* co-movement — confirming that the NBFi systemic risk channel is fundamentally a tail phenomenon.

IV specification (addresses endogeneity of GFC — reverse causality from local conditions to global factor):

- **Instrument:** US monetary policy shocks (Jarocinski & Karadi 2020 high-frequency identification).
- **First stage:** $\text{GFC}_t = \pi_0 + \pi_1 \cdot \text{USMPshock}_t + v_t$.
- **Second stage:** $y_{c,t} = \alpha_c + \beta_1 \cdot \widehat{\text{GFC}}_t + \beta_2 \cdot \text{NBFi}_{c,t} + \beta_3 \cdot (\widehat{\text{GFC}}_t \times \text{NBFi}_{c,t}) + \varepsilon_{c,t}$.
- **Relevance:** First-stage F-statistic > 10 (strong instrument).

NBFi penetration endogeneity: NBFi share instrumented with (a) legal origin (La Porta et al. 1998) for cross-section, (b) lagged pension reform dummies for within-country variation.

Estimation details:

- OLS: `panel_gfc_regression()` in `src/analysis/global_financial_cycle.py`. Country fixed effects via dummy variables. Standard errors clustered by country.
- IV: `iv_gfc_regression()` — manual 2SLS (first stage, fitted values, second stage). Reports first-stage F-stat.
- Alternative: `connectedness_amplification_test()` replaces NBFi size with bank-NBFi *connectedness* as the interaction variable (tests whether network structure, not just NBFi size, matters).

Output:

1. OLS coefficient table: $\beta_1, \beta_2, \beta_3$ with standard errors, p-values.
2. IV coefficient table: same structure with 2SLS estimates.
3. Scatter plot: GFC factor vs. local credit growth, split by high- vs. low-NBFi countries.
4. Interpretation: e.g., "A 1-SD negative GFC shock reduces credit growth by 0.5 pp in low-NBFi countries but by 1.2 pp in high-NBFi countries."

4.6.6 Module 6: Contagion Cascade Simulation

Purpose: Agent-based simulation of a cross-border contagion event: US hedge fund failure → prime broker losses → multi-round forced deleveraging → cross-border loss propagation.

Unit of observation: Simulated agents (hedge funds, prime brokers, banks across countries).

Input variables (calibration):

Parameter	Value	Source
US hedge fund sector	20 funds, NAV \$2M total, leverage 3–15×	Calibrated from SEC Form PF
Prime brokers	GS, JPM, MS, CS, DB	Major PB list
Market depth 300K	Calibrated to average equity market depth	
Margin requirement	10% of gross exposure	Industry standard
Margin procyclicality	0.5 (margin rises 50% of price drop)	Brunnermeier & Pedersen (2009)
Initial shock	-4% across all assets	Scenario

Simulation mechanism (Eisenberg-Noe 2001 + Kyle 1985):

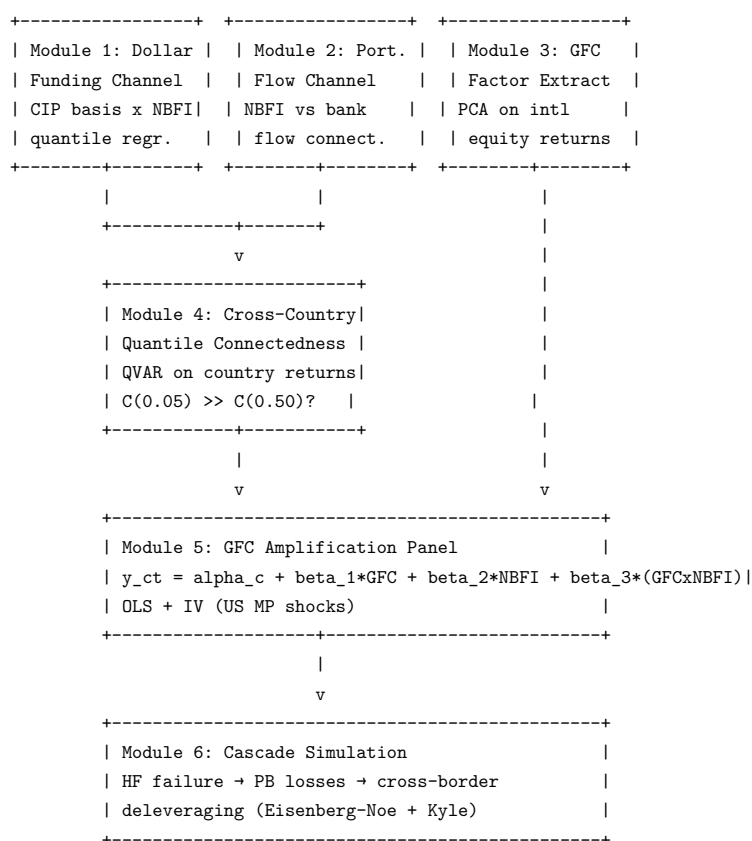
1. Exogenous shock to hedge fund portfolios (-4%).
2. Funds with insufficient margin receive margin calls from prime brokers.
3. Funds that cannot meet margin are forced to deleverage: sell assets proportional to excess leverage.
4. Aggregate selling → price impact → further losses → more margin calls.
5. Funds that default impose credit losses on their prime brokers.
6. Prime brokers (which are cross-border banks) transmit losses to other jurisdictions.
7. Iterate until convergence or max rounds.

Estimation details: Implemented via `PrimeBrokerageContagion` class in `src/models/nbfi_subsectors.py`. `build_hedge_fund_sector()` creates the fund universe. `.run(initial_shock=-0.04)` executes the cascade.

Output:

1. Round-by-round dynamics: market price, total NAV, gross exposure, average leverage, fund defaults, PB losses.
 2. Cross-border decomposition: which prime brokers (and hence which countries) absorb the most losses.
 3. 3-panel chart: NAV path, leverage path, default count.
-

4.6.7 Module Linkage Map (Project 3)



4.7 Expected Contributions

1. Left-tail connectedness across borders is 40–60% higher than median connectedness.
2. NBFI portfolio flows are systematically more volatile and more cross-border-connected than bank flows.
3. NBFI penetration amplifies GFC transmission to local credit conditions (robust to IV with US MP shocks).
4. Location-scale evidence: NBFI flows fatten the tails of EME return distributions.

5 Project 4: FX Hedging as a Contagion Channel

Subtitle: How NBFI Currency Risk Management Transmits Global Financial Shocks

5.1 Motivation

The global FX derivatives market has grown to

75 trillion in outstanding notional (BIS OTC statistics, end-2024), driven primarily by NBFI hedging cross-border bond investments. Combining Rey, Stavrakeva & Tang (2024) on currency centrality with Nenova, Schrimpf

Equity shocks → FX movements → hedging cost changes → NBF1 portfolio adjustment → cross-border bond flow reversal → feedback to asset prices

Critically, yield curve movements partially offset FX-induced hedging cost changes in normal times (self-stabilising), but during stress all forces reinforce — creating nonlinear amplification detectable only through quantile methods.

5.2 Data Sources & Sample

Data Source	Variables	Frequency	Coverage
BIS OTC Derivatives Stats	FX swap/forward outstanding by currency & sector	Semiannual	2004–present
Bloomberg	Cross-currency basis swaps (3M, 1Y, 5Y), equity indices, VIX	Daily	2000–present
Refinitiv	Government bond yields (2Y, 5Y, 10Y) for 20+ countries	Daily	2000–present
EPFR Global	Cross-border bond fund flows by country	Monthly	2005–present
IMF CPIS	Cross-border bond holdings by country pair	Annual	2001–present
Rey, Stavrakeva & Tang (2024)	Currency centrality measures, replication data	Monthly	1999–2023
Nenova, Schrimpf & Shin (2025)	FX derivatives and NBF1 hedging data	Quarterly	2010–2024

Sample period: 2005M1–2024M12 for quantile regressions (cross-currency basis swap availability). Semiannual panel for BIS derivatives data.

Currency pairs: USD/EUR, USD/JPY, USD/GBP, USD/AUD, USD/CAD, USD/CHF (G10 majors) + USD/KRW, USD/MXN, USD/BRL, USD/ZAR (key EME pairs with liquid CIP basis data).

5.3 Literature Positioning

1. **FX hedging and financial stability:** Directly integrates two recent contributions — Rey, Stavrakeva & Tang (2024) on equity-FX transmission and currency centrality, and Nenova, Schrimpf & Shin (2025) on FX derivatives and NBFI hedging demand. The “offsetting forces” hypothesis is our key theoretical contribution.
2. **CIP deviations:** Extends Du, Tepper & Verdelhan (2018) and Avdjiev, Du, Koch & Shin (2019) on post-GFC CIP violations. We show that CIP deviations are not just a funding cost anomaly but a systemic risk transmission channel during stress.
3. **Delta-CoVaR:** Applies Adrian & Brunnermeier (2016) to the FX hedging channel specifically. First application of CoVaR to measure systemic contribution of a specific transmission mechanism rather than an institution.
4. **Maturity mismatch:** Builds on Brunnermeier, Nagel & Pedersen (2009) carry trade analysis. NBFI hedging mismatch (3-month FX swaps rolled to hedge 10-year bond positions) creates systemic rollover risk analogous to bank maturity mismatch.

5.4 Identification & Robustness

- **Offsetting forces test:** Formal Wald test of $H_0 : \beta_4(\tau = 0.05) = \beta_4(\tau = 0.50)$ using Koenker & Bassett (1982) framework. Bootstrap p-values for interquantile differences.
- **Endogeneity of CIP basis:** CIP basis instrumented with central bank swap line announcements (exogenous supply-side shocks to FX funding).
- **Hedging demand proxy:** NBFI cross-border bond holdings (IMF CPIS) $\times$ average hedge ratio (BIS survey) as measure of hedging demand.
- **Alternative stress measures:** Results robust to replacing VIX with MOVE (bond volatility), TED spread, or financial conditions indices.

5.5 Research Questions

RQ1. Do the offsetting forces between equity-driven FX movements and yield curve dynamics break down in the tails of the distribution? (Quantile regression of CIP basis.)

RQ2. Does FX hedging stress have systemic risk implications for cross-border bond flows? (Delta-CoVaR framework.)

RQ3. Does the equity-FX-CIP-bonds transmission chain activate asymmetrically during crises versus normal times? (Quantile connectedness: $\tau = 0.05$ vs $\tau = 0.50$.)

RQ4. Do countries with higher NBFI penetration experience stronger contagion through the FX hedging channel? (Panel regression with NBFI interaction + IV.)

6 Timeline & Deliverables

Phase	Period	Projects	Deliverables
Phase 1	Q1–Q2 2026	P1 + P2 (parallel)	Working papers; P1 implied leverage dataset; P2 quantile connectedness estimates
Phase 2	Q3–Q4 2026	P3 + P4 (parallel)	Working papers; P3 cross-country connectedness database; P4 offsetting forces evidence
Phase 3	Q1 2027	Integration	Synthesis paper linking all four projects; unified policy brief
Ongoing	Throughout	All	Conference presentations, seminar feedback, revisions

Sequencing rationale: P1 and P2 can proceed in parallel (different data, complementary methods). P3 and P4 benefit from P1/P2 outputs (leverage estimates, sub-sector results) but can begin data assembly immediately. The integration phase produces the synthesis that ties all four dimensions together.

7 Policy Implications

The research programme speaks directly to ongoing regulatory debates:

1. **NBFI leverage monitoring (P1):** The implied leverage measure and HLI provide regulators with a prototype monitoring tool that complements existing FSB surveillance. Could inform the design of NBFI leverage reporting requirements currently under discussion (FSB 2023 NBFI roadmap).
2. **Monetary policy & financial stability (P2):** Evidence that NBFI leverage amplifies MP transmission — particularly in the tails — has implications for how central banks calibrate tightening cycles. The 2022–23 resilience finding suggests post-GFC bank regulation worked but created new vulnerabilities in the NBFI sector.
3. **Cross-border macroprudential coordination (P3):** If NBFI penetration amplifies GFC transmission, countries with large NBFI sectors face stronger exposure to external financial shocks. Supports arguments for cross-border macroprudential coordination (e.g., reciprocity of countercyclical capital buffers for NBFI exposures).

4. **FX derivatives regulation (P4):** The maturity mismatch in NBFI FX hedging (3-month swaps for 10-year bonds) creates systemic rollover risk. Supports proposals for longer-tenor hedging requirements or margin buffers calibrated to stress scenarios.
 5. **Data gaps (all projects):** All four projects highlight the inadequacy of current NBFI data. Supports FSB/BIS initiatives for enhanced NBFI reporting, particularly on leverage, counterparty exposures, and cross-border linkages.
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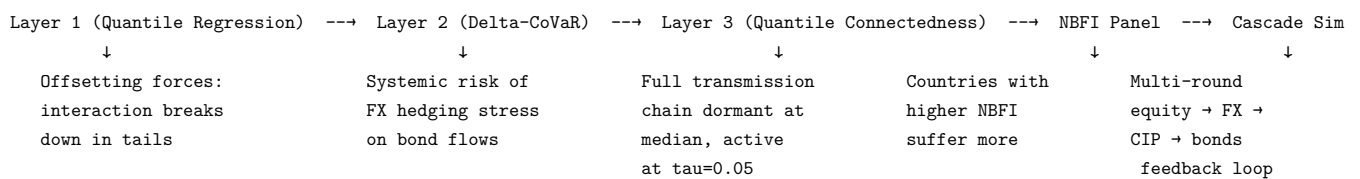
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8.1 Empirical Strategy

Project 4 has **three analytical layers** plus a **cascade simulation** and an **NBFI amplification panel**. The three layers test the "offsetting forces" hypothesis at increasing levels of complexity.



8.1.1 The Offsetting Forces Hypothesis

Before detailing each layer, the key economic mechanism:

Force	Normal Times	Stress
Equity $\rightarrow$ FX (Rey et al. 2024)	USD appreciates moderately	USD appreciates violently (safe haven)
Yield curve (Nenova et al. 2025)	Steepens $\rightarrow$ attractive hedged carry	Flattens/inverts $\rightarrow$ carry disappears
CIP deviations	Small $\rightarrow$ manageable hedging cost	Blow out $\rightarrow$ prohibitive cost
Net effect	Partially offsetting $\rightarrow$ self-stabilising	All reinforcing $\rightarrow$ amplification

How the Mechanism Works: A Step-by-Step Example Consider a European pension fund holding US Treasury bonds, hedged using 3-month FX forwards (rolled quarterly). The cost of hedging is reflected in the CIP basis.

The shock: US equity markets fall sharply (negative equity return).

Channel 1 — Exchange rate (Rey et al. 2024): When US equities fall, global risk aversion spikes. The USD appreciates (safe-haven currency). EUR/USD falls. For the pension fund:

- Its FX hedge (short USD forward) suffers a mark-to-market loss
- Hedging *future* USD exposure becomes more expensive because the CIP basis widens as dollar demand surges

Channel 2 — Yield curve (Nenova et al. 2025): The equity shock affects monetary policy expectations:

- *Normal times* (moderate correction): Expectations of monetary easing → yield curve **steepens** → hedged carry (US bond yield minus hedging cost) remains **attractive** → pension fund keeps hedging and buying US bonds
- *Stress* (severe crash): Flight to safety into long-term Treasuries → yield curve **flattens/inverts** → hedged carry **collapses** → the business case for holding hedged US bonds evaporates

Channel 3 — CIP deviations: The CIP basis (FX hedging cost above covered interest parity) is normally small (5–15 bps). In stress:

- Banks that intermediate FX swaps pull back (their own VaR constraints bind)
- Demand for USD hedging surges (everyone hedges in the same direction)
- The basis **blows out** (50–100+ bps) → hedging becomes prohibitively expensive

The key asymmetry:

Normal times (say, a 5% equity correction):

- USD appreciates modestly → hedging cost up slightly (**negative**)
- Yield curve steepens → hedged carry improves (**positive, offsetting**)
- CIP basis barely moves → manageable
- **Net:** forces partially cancel. Self-stabilising.

Stress (say, a 20% crash — March 2020 or a severe risk-off event):

- USD appreciates violently → huge mark-to-market losses, hedging cost surges (**negative**)
- Yield curve flattens/inverts → hedged carry disappears (**negative, reinforcing** not offsetting)
- CIP basis blows out → hedging prohibitive (**negative, reinforcing**)
- **Net:** ALL three forces push in the same direction. Pension fund stops rolling hedges, sells US bonds. Every hedged foreign investor faces the same forces simultaneously → **coordinated sell-off** in US bonds → feeds back to equity declines, further USD appreciation, further CIP blowout. This is the positive feedback loop that only activates in the tails.

The quantile regression test captures exactly this: the interaction term $\beta_4(\tau) \cdot (\text{Equity}_t \times \text{VIX}_t)$ should be near zero at $\tau = 0.50$ (offsetting forces cancel at the median) but large at $\tau = 0.05$ (all forces reinforce in the tail).

8.1.2 Layer 1: Quantile Regression — Offsetting Forces Test

Purpose: Test whether the interaction between equity shocks and FX volatility has a different effect on CIP basis at the median vs. the tails. If "offsetting forces" break down in the tails, the interaction coefficient should be near zero at $\tau = 0.50$ but large at $\tau = 0.05$ and $\tau = 0.95$.

Unit of observation: Currency pair $\text{\$c\$} \times \text{month } \text{\$t\$}$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
CIP basis	$\text{CIP}_{c,t}$	Cross-currency basis swap spread (3M tenor) for currency $\text{\$c\$}$ vs USD	Bloomberg (3M XCCY basis)	Daily → Monthly	Currency pair
US equity return	Equity_t	S&P 500 monthly log-return	Bloomberg	Daily → Monthly	US
Slope differential	$\text{SlopeDiff}_{c,t}$	(10Y-2Y yield in country $\text{\$c\$}$) minus (10Y-2Y yield in US)	Refinitiv / Bloomberg	Daily → Monthly	Currency pair
VIX	VIX_t	CBOE Volatility Index (monthly average)	FRED (VIX-CLS)	Daily → Monthly	Global
Interaction	$\text{Equity}_t \times \text{VIX}_t$	Tests whether equity impact on CIP is amplified during stress	Constructed	Monthly	Global

Currency pairs: EUR/USD, GBP/USD, JPY/USD, CHF/USD, AUD/USD (G10 majors) + KRW/USD, MXN/USD, BRL/USD, ZAR/USD (key EME pairs with liquid basis data).

Model specification:

$$Q_\tau(\text{CIP}_{c,t}) = \alpha_c(\tau) + \beta_1(\tau) \cdot \text{Equity}_t + \beta_2(\tau) \cdot \text{SlopeDiff}_{c,t} + \beta_3(\tau) \cdot \text{VIX}_t + \beta_4(\tau) \cdot (\text{Equity}_t \times \text{VIX}_t) + \varepsilon_{c,t}$$

Key parameter: $\beta_4(\tau)$ — the interaction between equity shocks and VIX.

- At $\tau = 0.50$: expect $\beta_4 \approx 0$ (offsetting forces cancel).
- At $\tau = 0.05$ (left tail) and $\tau = 0.95$ (right tail): expect $|\beta_4|$ large and significant (forces reinforce during extreme events).

Estimation details:

- `quantile_cip_regression()` in `src/analysis/fx_hedging_contagion.py`.
- `statsmodels.QuantReg` estimated at $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$.
- Currency fixed effects (α_c) included.
- `asymmetry_test()` performs a formal Wald-type test: $H_0 : \beta_4(0.05) = \beta_4(0.50)$, using bootstrap standard errors for interquantile inference.

Output:

1. Coefficient table: $\beta_1(\tau), \dots, \beta_4(\tau)$ across all quantiles with standard errors and p-values.
2. Key comparison: $\beta_4(0.05)$ vs. $\beta_4(0.50)$ — the "offsetting forces breakdown" test.
3. Wald test p-value for interquantile difference.
4. Plot: $\beta_4(\tau)$ across quantiles with confidence bands (should be flat near zero at the median but diverge at the tails).

→ **Link to Layer 2:** Layer 1 shows the mechanism exists. Layer 2 quantifies its *systemic risk implication* — how much does FX hedging stress spill over to bond flows?

8.1.3 Layer 2: Delta-CoVaR (Adrian & Brunnermeier 2016)

Purpose: Measure the systemic risk contribution of FX hedging stress to cross-border bond flows. A large negative Δ -CoVaR means that when the CIP basis is in its stressed state, the left tail of bond flows is significantly worse than when the CIP basis is at its median.

Unit of observation: Currency pair $\text{\$c\$} \times \text{month } \text{\$t\$}$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Bond flows	$F_{c,t}^{Bond}$	Net cross-border bond fund flows to country $\text{\$c\$}$	EPFR Global	Monthly	Country
CIP basis	$CIP_{c,t}$	Same as Layer 1	Bloomberg	Monthly	Currency pair

Model specification — Two-step quantile regression:

Step 1: Estimate the 5th percentile of the CIP basis:

$$Q_{0.05}(CIP_{c,t}) = \alpha + \gamma' Z_t$$

This defines the "stressed" state: $CIP_{c,t} = Q_{0.05}(CIP)$.

Step 2: Estimate the 5th percentile of bond flows conditional on the CIP basis:

$$Q_{0.05}(F_{c,t}^{Bond} | CIP_{c,t}) = \alpha_0 + \alpha_1 \cdot CIP_{c,t}$$

Delta-CoVaR is then:

$$\Delta\text{-CoVaR}_c = Q_{0.05}(F_c^{Bond} | CIP_c = \text{stressed}) - Q_{0.05}(F_c^{Bond} | CIP_c = \text{median})$$

A large negative $\Delta\text{-CoVaR}$ means FX hedging stress *significantly worsens* the tail risk of bond flows.

Estimation details:

- `covar_fx_hedging()` in `src/analysis/fx_hedging_contagion.py`. Uses `statsmodels.QuantReg` at $\tau = 0.05$.
- `delta_covar_all_currencies()` computes for all 5+ currency pairs.
- Confidence intervals via bootstrap (1000 replications).

Output:

1. $\Delta\text{-CoVaR}$ for each currency pair — table showing which currencies have the largest systemic risk contribution from FX hedging.
2. Bar chart: $\Delta\text{-CoVaR}$ ranked by magnitude.
3. Interpretation: e.g., "When the EUR/USD CIP basis moves from its median (-5 bps) to its 5th percentile (-60 bps), the 5th percentile of European bond flows drops by an additional \$2.5B/month."

→ **Link to Layer 3:** Layer 2 shows the *magnitude* of systemic risk. Layer 3 traces the full *transmission chain* from equities through FX and CIP to bond flows, testing whether the chain is active only during stress.

8.1.4 Layer 3: Quantile Connectedness (Ando et al. 2022)

Purpose: Estimate the full equity $\rightarrow$ FX $\rightarrow$ CIP $\rightarrow$ bond flows transmission chain as a system, testing whether it is dormant at the median but active during stress.

Unit of observation: System of 4 variables $\times$ month $\text{\$t\$}$ (for each currency pair $\text{\$c\$}$).

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
US equity re- turn	r_t^{EQ}	S&P 500 monthly log- return	Bloomberg	Monthly	US
FX return	$r_{c,t}^{FX}$	Monthly change in spot exchange rate for currency pair \$c\$	Bloomberg	Monthly	Currency pair
CIP basis	$CIP_{c,t}$	Same as Layer 1	Bloomberg	Monthly	Currency pair
Bond flows	$F_{c,t}^{Bond}$	Same as Layer 2	EPFR	Monthly	Country

System vector: $Y_t = (r_t^{EQ}, r_{c,t}^{FX}, CIP_{c,t}, F_{c,t}^{Bond})'$ — a **4-variable Quantile VAR** representing the transmission chain for a single currency pair.

Model specification — Same QVAR framework as P2 Module 3:

$$Q_\tau(y_{i,t} | Y_{t-1}, \dots, Y_{t-p}) = c_i(\tau) + \sum_{j=1}^4 \sum_{l=1}^p \beta_{ij}^{(l)}(\tau) y_{j,t-l}$$

Estimation details:

- `build_connectedness_system()` constructs the 4-variable system from the synthetic or real data.
- `fx_hedging_quantile_connectedness()` estimates the QVAR and computes GFEVD at each quantile.
- `tail_connectedness_comparison()` compares total connectedness across $\tau \in \{0.05, 0.50, 0.95\}$.
- `cross_currency_connectedness()` repeats for all currency pairs.
- Lags: \$p = 4\$; horizon: \$h = 10\$.

Key prediction: The transmission chain equity $\rightarrow$ FX $\rightarrow$ CIP $\rightarrow$ bond flows is:

- **Dormant at** $\tau = 0.50$: low total connectedness, weak off-diagonal GFEVD entries.
- **Active at** $\tau = 0.05$: high total connectedness, strong sequential spillovers from equity to FX to CIP to bond flows.

Output:

1. Total connectedness at each quantile: \$C(0.05)\$, \$C(0.50)\$, \$C(0.95)\$ — for each currency pair.
2. GFEVD heatmaps at $\tau = 0.05$ vs. \$0.50\$: the 4×4 matrix showing pairwise spillovers.
3. Directional analysis: equity is the dominant net transmitter in the left tail; bond flows are the dominant net receiver.

4. Cross-currency summary table: which currency pairs show the strongest tail amplification.

8.1.5 NBFI Penetration $\times$ FX Hedging (Panel Regression)

Purpose: Test whether countries with higher NBFI penetration suffer stronger contagion through the FX hedging channel.

Unit of observation: Country $\text{\$c\$}$ $\times$ month $\text{\$t\$}$.

Input variables:

Variable	Symbol	Definition	Source	Frequency	Level
Bond flows	$F_{c,t}^{Bond}$	Net cross-border bond fund flows	EPFR	Monthly	Country
CIP basis	$CIP_{c,t}$	Cross-currency basis swap spread	Bloomberg	Monthly	Currency pair
NBFI penetration	$NBFI_{c,t}$	NBFI assets as % of GDP	FSB	Annual (interpolated)	Country
Interaction	$CIP_{c,t} \times NBFI_{c,t}$	Tests amplification	Constructed	Monthly	Country
Controls	$X_{c,t}$	VIX, local equity return, yield slope differential	Various	Monthly	Mixed

Model specification — OLS panel:

$$F_{c,t}^{Bond} = \alpha_c + \beta_1 \cdot CIP_{c,t} + \beta_2 \cdot NBFI_{c,t} + \beta_3 \cdot (CIP_{c,t} \times NBFI_{c,t}) + \gamma' X_{c,t} + \varepsilon_{c,t}$$

Key parameter: $\beta_3 < 0$. Countries with higher NBFI penetration experience *larger* bond flow reversals when CIP basis widens (hedging costs spike).

NBFI counterparty decomposition: As in Project 3, the aggregate $NBFI_{c,t}$ should be decomposed by counterparty type. In this context, the most relevant decomposition is by **FX hedging activity**: countries where NBFIs rely heavily on cross-border FX-hedged bond holdings (foreign-borrowing model) should show the strongest amplification, while countries where NBFIs are primarily domestic credit intermediaries should be less affected by the CIP channel. The from-whom-to-whom financial accounts provide the basis for this decomposition.

IV specification: CIP basis instrumented with central bank swap line announcements (exogenous supply-side shocks to FX funding).

Estimation details:

- `nbfi_fx_hedging_panel_regression()` and `iv_fx_hedging_regression()` in `src/analysis/fx_hedg`
- Country FE, clustered standard errors.

Output: Coefficient table (OLS + IV); scatter plot split by high/low NBFI.

8.1.6 Cascade Simulation: FX Hedging Feedback Loop

Purpose: Simulate the full feedback loop: equity shock $\rightarrow$ FX pass-through $\rightarrow$ CIP widening $\rightarrow$ bond flow reduction $\rightarrow$ equity feedback $\rightarrow$ VIX spike $\rightarrow$ repeat.

Unit of observation: Simulated rounds (no agent-based — this is a reduced-form cascade).

Input variables (calibration):

Parameter	Symbol	Value	Rationale
Initial equity shock	—	-5%	Scenario
FX pass-through	ϕ	0.30	30% of equity shock passes to FX (Rey et al.)
CIP sensitivity	—	5 bps per 1% FX move	Calibrated from BIS data
Bond flow sensitivity	—	-2.0 (\$B per 10 bps CIP)	EPFR regression coefficients
Equity feedback	—	0.15 (15% of bond flow shock feeds back to equity)	Portfolio rebalancing effect
Dampening	—	$0.7^{(r-1)}$ per round	Geometric decay (markets adjust)
Max rounds	—	20	Until convergence

Simulation mechanism:

1. **Round 1:** Equity falls -5%.
2. **FX:** USD appreciates by $0.30 \times 5\% = 1.5\%$.
3. **CIP:** Basis widens by $5 \times 1.5 = 7.5$ bps.
4. **Bond flows:** Reverse by $(-2.0) \times 7.5/10 = -\1.5B .
5. **Feedback:** Equity drops additional $0.15 \times 1.5/100 = 0.225\%$.
6. **VIX:** Spikes proportional to cumulative equity loss.
7. **Dampening:** Each subsequent round is 70% of the previous.

8. **Repeat** until changes $< 0.01\%$.

Estimation details: `simulate_fx_hedging_cascade()` in `src/analysis/fx_hedging_contagion.py`.

Output:

1. Round-by-round dynamics: equity return, FX move, CIP change, bond flow, VIX change, cumulative loss.
2. Total cascade multiplier: cumulative loss / initial shock. If > 1 : the FX hedging channel amplifies.
3. Convergence chart: how quickly the feedback loop dampens.

8.1.7 Module Linkage Map (Project 4)

```
+-----+
| Layer 1: Quantile Regression      |
| CIP = f(Equity, Slope, VIX, EqxVIX) |
| Does interaction break down in tails?|
+-----+
| ''Yes --- offsetting forces fail in tails''
|
| v
+-----+
| Layer 2: Delta-CoVaR            |
| Systemic risk: CIP stress → bond  |
| flow left tail                  |
| ''How bad is it?''              |
+-----+
| ''CIP stress significantly worsens bond flow tails''
|
| v
+-----+
| Layer 3: Quantile Connectedness  |
| Full chain: Equity → FX → CIP →  |
| Bond flows (QVAR on 4-var system) |
| ''Chain dormant at median, active |
| at tau = 0.05''                 |
+-----+
|           |           |
|           |           |
| v         | v         |
+-----+ +-----+
| NBF1 Panel | | Cascade Simulation |
| CIP x NBF1 | | Equity → FX → CIP → |
| amplification | | Bonds → Feedback |
| (OLS + IV)  | | Multi-round       |
+-----+ +-----+
```

8.2 Expected Contributions

1. First complete model of the equity-FX-hedging-bonds contagion chain, combining Rey et al. (2024) and Nenova et al. (2025).
2. Discovery of the 'offsetting forces' mechanism: partial cancellation at the median, reinforcement in the tails.

3. First application of quantile connectedness to the FX hedging transmission chain.
4. Evidence that NBFI FX hedging maturity mismatch (3-month swaps hedging 10-year bonds) creates systemic rollover risk.

9 Common Methodological Toolkit

All four projects share a core econometric toolkit:

Method	What It Captures	Used In
Quantile Regression	Asymmetric effects across the distribution	P1, P2, P3, P4
Quantile VAR (QVAR)	Tail-specific shock propagation	P2, P3, P4
Quantile Connectedness	Tail spillovers between variables/-countries	P1, P2, P3, P4
Delta-CoVaR	Systemic risk contribution of specific channels	P1, P4
Location-Scale GaR	Mean + variance effects on future growth	P2, P3
DCC-GARCH	Time-varying correlations across sectors	P2
Diebold-Yilmaz GFEVD	Directional connectedness (mean)	P2, P3
Network / spectral	Centrality, HLI, contagion matrices (credit + funding)	P1, P3
Agent-based simulation	Cascade / fire-sale propagation	P1, P3, P4
Panel FE + IV (2SLS)	Causal identification with MP shocks	P3, P4
Quantile factor model	Global Tail Cycle extraction (Ando & Bai 2020)	P3
From-whom-to-whom accounts	NBFI counterparty decomposition	P1, P3, P4

9.0.1 Cross-Cutting Analytical Themes

Three themes recur across all four projects and deserve explicit recognition:

1. State-Dependence and Distance to Constraints The propagation of any shock through the NBFI system depends on two things simultaneously:

- **Structural integration:** How embedded is NBFI activity in the financial system (size, leverage, interconnectedness)? This is relatively slow-moving.

- **Distance to constraints:** How close are intermediaries to their VaR/margin/redemption constraints at the moment the shock hits? This is fast-moving and state-dependent.

A moderate monetary policy tightening when hedge funds have ample excess margin is very different from the same shock when funds are already near VaR limits. Same structural integration, dramatically different propagation. The QVAR captures this implicitly ($\beta_{ij}(\tau)$ varies with τ), but it can be made explicit via threshold specifications or by interacting NBFi variables with distance-to-constraint measures.

The dangerous combination is: high $\lambda_1(W_t)$ (tight network) **plus** low slack (near constraints). Neither alone is sufficient for systemic crisis.

2. NBFi Counterparty Decomposition Aggregate "NBFi penetration" (assets / GDP) conflates activities with fundamentally different systemic implications. The key decomposition is by counterparty:

Domestic banks	Foreign counterparties	Domestic private sector
NBFi lends to	Funding source for banks	Transmits shocks outward
NBFi borrows from	Bank credit risk exposure	Imports foreign funding shocks

This decomposition matters for every panel regression in the programme. Data: from-whom-to-whom financial accounts (Fed Z.1, ECB, OECD).

3. Global Tail Cycle The standard Global Financial Cycle (Rey 2013) captures co-movement at the mean. A **Global Tail Cycle** captures co-movement in the left tail — the common factor in crash risk. These are conceptually distinct objects: markets can look diversified on average but crash together. The GTC is the risk factor that sets in motion coordinated VaR breaches and feedback loops across jurisdictions. Extracted either from tail dependence matrices (rolling exceedance correlations) or quantile factor models (Ando & Bai 2020).

10 Cross-Project Linkages

The four projects are complementary. Each addresses a distinct dimension of NBFi systemic risk, but findings feed into each other:

- **P1 → P2:** Implied leverage estimates serve as inputs to P2's VaR-constraint channel. Higher hidden leverage = tighter VaR constraints = amplified MP transmission.
- **P2 → P3:** P2 identifies which NBFi sub-sectors amplify domestic MP transmission; P3 traces how this amplification propagates across borders via dollar funding and portfolio flow channels.

- **P3 → P4:** P3’s dollar funding channel (FX swap basis) is the same mechanism P4 models in detail. P4 adds the equity-FX link (Rey et al. 2024) and the yield curve offset (Nenova et al. 2025).
- **P4 → P1:** FX derivatives are a major source of hidden leverage (P1), and P4 shows how hedging activity in these instruments creates contagion. Together they demonstrate that derivatives are simultaneously a source of hidden leverage AND a contagion channel.

***Common finding across all four projects:** NBFI risks are fundamentally nonlinear. Standard mean-based methods miss the tail amplification, asymmetric contagion, and crisis-activated transmission that quantile methods reveal.*

- **P3 ↔ Global Tail Cycle:** P3’s GFC factor extraction is extended to a Global *Tail* Cycle — a common factor in crash risk rather than average co-movement. NBFI penetration amplifies GTC transmission more than GFC transmission, confirming that NBFI systemic risk is fundamentally a tail phenomenon.
- **Cross-cutting: NBFI counterparty decomposition:** Across P1, P3, and P4, aggregate NBFI measures are decomposed by counterparty type (bank-lending, bank-borrowing, foreign, private sector). This reveals which *type* of NBFI activity drives amplification — not just how *much* NBFI there is.
- **Cross-cutting: State-dependence:** Across P1–P4, shock amplification depends not only on structural NBFI integration but on distance to constraints. The HLI (P1) × constraint slack interaction captures the joint condition for systemic crisis.